

Great Expectations: Responses to Current and Future Transfers for Low-Income Individuals*

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Abstract

How does the expectation of aid change behavior? We propose a simple approach to separate expectations effects from the direct effects of relaxing resource constraints: compare the promise of a program to the program itself. We test this with respect to cash transfers in a four-arm randomized controlled trial in Uganda. Both those who received and those promised-to-receive increase their labor supply and investment. Immediate transfers also increase household expenditures and savings. Our results are not consistent with standard life-cycle models; they are better explained by a model in which the transfer increases individual labor productivity.

Keywords: expectations, cash transfers, life-cycle model, aid dependency, labor supply, work disincentives, Uganda

JEL: D15, J22, O12

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1 Introduction

To understand the determinants and consequences of persistent poverty, economists often study the causal impacts of policy interventions. Naturally, few if any interventions perfectly isolate one constraint. In some cases studying the effects of a credible promise of future intervention may provide insights complementary to the examination of the intervention itself. Policy interventions – and transfers to households in particular – may shift expectations that affect current actions, such as investment, consumption, and labor supply decisions. Such interventions also often involve a stream of current and future transfers, thus shifting immediate liquidity, future liquidity, and, critically to what we study, expectations of future liquidity. The estimated treatment effects of such interventions on economic outcomes inevitably reflect a difficult-to-separate combination of direct consequences of the relaxed liquidity constraint and the indirect consequences of expectation shifts.

We separate these two effects in the context of cash transfers in a randomized controlled trial in Uganda. Our design allows us to test whether the expectation of future income, in the form of a cash transfer, changes behavior. Theories abound that predict both positive and negative impacts of cash transfers on labor supply and investment in income-generating activities. Canonical life-cycle models of the labor-leisure tradeoff (Deaton, 1991; Hall, 1978) predict that the provision of aid should reduce labor supply all else equal. Yet recent studies from low- and middle-income countries consistently produce null or positive effects on labor supply (Banerjee et al., 2017; Crosta et al., 2025).¹

Models explaining the null or positive pattern have focused on two potential channels: cash transfers might either relax liquidity constraints, allowing for lumpy productive investments, or alternatively cash transfers may impact individual productivity directly through physical, mental, or psychological wellbeing – for example, by providing peace of mind via a buffer-stock or increased aspirations which then motivates increased investment. Distinguishing between these mechanisms has important implications for our understanding of how low-income individuals make economic choices, which is critical for the design of social protection policy. If the latter mechanism is at play, changing expectations (alone) could have important impacts on labor supply, self-employment income, and wealth accumulation.

Our experimental design explicitly separates the expectation of future aid from the receipt of current transfers. An *Expectations* treatment creates the expectation of future aid by promising beneficiaries a large unconditional lump-sum cash transfer (US\$135 nominal in 2013; equivalent to two months of average household income) to be delivered in one year’s time. We compare

¹Although results may differ in wealthier countries: in two recent studies in the United States, transfers reduced labor supply for the full sample in Vivalt et al. (2025) and for the sub-sample that engaged in part-time work in Balakrishnan et al. (2025).

the outcomes of this group to a pure control group (no cash, *Control*), as well as two other treatment groups that received the same transfer either immediately (*Cash*), or immediately but after completing two brief budget planning sessions (*Cash with Planning*).

Our findings amount to a repudiation of the aid dependency argument, at least in the study context. The promise of a future transfer (*Expectations*) caused an immediate *increase* in work hours and business profits, as well as greater consumption of nutritious food. In the short term, the anticipated transfer did not reduce saving – and, if anything, it decreased debt, counter to the idea that beneficiaries might borrow on the “collateral” created by the promise of future aid. Both the *Cash* and *Cash with Planning* treatments had even larger impacts, generating immediate increases in labor supply, business investment, saving, and nutritive consumption, similar to the findings of previous studies. At endline, just prior to the *Expectations* group receiving cash, labor supply, food security and dietary diversity, household expenditures, and durables were all significantly higher in the *Expectations* arm relative to *Control*. Effects on these outcomes at one year for the *Cash* and *Cash with Planning* arms are substantially positive as well, and about twice as large as for *Expectations*. Taken together, the results show that the expectation of future aid elicited positive labor supply and business productivity responses and led to improvements in important measures of well-being even before cash was delivered to beneficiaries.

What explains these effects? And how should we revise our thinking on the way individuals experiencing poverty make decisions concerning labor supply and entrepreneurship? Several natural extensions of the canonical model fail to explain what we see in the data. These include models featuring credit market imperfections à la Deaton (1991), as well as lumpy durables models in the vein of Banerjee et al. (2015). We find that the model most consistent with our results is one in which transfers affect labor productivity.

This effect may arise through a variety of mechanisms. First, mental health and productivity are closely related. Recent work has shown that depression and anxiety – as well as hope, aspirations, and other components of psychological well-being – are key causal determinants of productivity and economic success (Banerjee et al., 2025; Kaur et al., 2025; Lund et al., 2024). The expectation of future transfers could affect current-period mental health through various channels (e.g., by reducing depression or raising aspirations), thus expanding the capacity for effort, resulting in an increase in labor supply.

Second, prior studies have shown the importance of risk in limiting low-income individuals’ investments in productive activities. Perhaps the utility cost of losing whatever small amount of liquidity individuals had at baseline is too large, and larger investments would put that initial liquidity at risk too. This would create a disincentive to launch a new business endeavor when labor and capital are complements in the production function, even if the liquidity constraint is not binding. The promise of future aid changes this calculus by essentially insuring against business

failure, and thus incentivizes greater labor input and other business investments that individuals would otherwise not have risked making.

Third, while most models of financing for the poor focus on helping to meet the large threshold requirements of starting businesses, in reality many entrepreneurial projects, especially the micro-enterprises in which low-income individuals often engage, have small start-up and learning costs (imagine, for example, selling fruit by the roadside or weaving straw baskets for sale in the local market). Larger capital expenses kick in only if the enterprise expands considerably (e.g., renting a storefront or buying heavy machinery). The promise of a transfer could induce individuals to embark on such projects, starting small with available capital and effort, but knowing that should the business grow they will have the ability to make bigger investments in the future. While we have some evidence to bring to bear upon these ideas, data limitations restrict our ability to test the relative importance of each. We are not able in our context and data to validate or refute the differential relevance of these three mechanisms.

Our main contribution is to propose a novel and easily replicable way to disentangle expectations effects from the direct effects of interventions *per se*. The impacts of most safety net policies reflect a combination of these effects; separating expectations effects enables a deeper understanding of the ways in which such policies affect current and future economic decision-making. We demonstrate through our results that life-cycle models of labor, consumption, and investment for the poor should likely feature a direct role for transfers in augmenting labor productivity, yielding very different predictions of behavioral responses to safety net policies than standard models.

Our work also complements a growing body of evidence on the effects of interventions designed to change expectations directly (often construed as aspirations, though such interventions may shift expectations either in levels or in beliefs about the riskiness of future outcomes (Jensen, 2010; Karlan et al., 2014)). While much of this work has been focused on education and household investments in children, Orkin et al. (2014) show that unrestricted cash grants raise aspirations, and that the impacts of cash are broadly similar to the impacts of a video intervention designed to impact aspirations directly. We build on these findings by examining the role of timing, showing that merely anticipating a cash transfer leads to changes in labor supply and consumption similar to those observed when aspirations are directly manipulated.

Lastly our work is a cautionary tale for randomized phase-in design program evaluations to not pre-announce to all groups what they will later receive (or, perhaps better, to study what the consequence is of the pre-announcement). This issue is similar to the program evaluation challenge pointed out long ago that behavior often changes immediately prior to the commencement of a program (see Ashenfelter, 1978, Ashenfelter and Card, 1985; and Deshpande and Dizon-Ross, 2023, for a more recent example). Usually due to logistical considerations and ethical concerns, many evaluations employ a strategy that promises the treatment with a delay to the control group. This

strategy presumes that the promise of treatment does not meaningfully change behavior within the measurement period. While the practical merits of this strategy are clear, we find that conflating delayed treatment and pure control would result in a biased treatment effect estimate, specifically an underestimate of the treatment effect (although obviously the expectation effect, in other situations, could push against rather than in the same direction of the treatment effect intended by the program designers). Thus our findings underscore the critical importance of taking expectation impacts into account when designing evaluations of social safety net policies.²

2 Research Design and Data

We partnered with The AIDS Support Organization (TASO), a clinical delivery and support organization based in Uganda that provides care to over 100,000 Ugandans with HIV and their families. TASO operates throughout Uganda, with 54 public health facilities and 11 regional centers across the country (Bakanda et al., 2011a,b; Chu et al., 2013; Mills et al., 2011). Patients receive antiretroviral therapy (ART) and are also offered voluntary monthly counseling sessions to help them and their families cope with the illness. TASO was primarily interested in examining whether cash transfers would lead to a higher CD4⁺ T-cell count, a biomarker tracked for HIV patients. Mills et al. (2018) reports null results on that, along with mostly null but positive treatment effects on other physical health outcomes.

Participants were randomly assigned to one of four experimental arms with stratification by TASO center (Masindi or Soroti), gender, and age.³ The four arms were as follows:

Transfer (T1): Individuals assigned to T1 were informed that they had been selected to receive a cash grant to improve their overall welfare, to spend as they wish, and that they would receive the money at their next monthly counseling session. They received no guidance on how they could or should spend the grant.

Transfer Plus Planning (T2): Individuals assigned to T2 were told that they had been selected to receive a cash grant at their next monthly counseling session, and that the grant was intended to improve their overall welfare, to spend as they wish. However, prior to receiving the transfer, they were required to attend two financial planning sessions held one week apart. These sessions provided information on how recipients *could* spend their transfer, and discussed the temptations and social pressure to share that they might face when they received the money. Individuals were then

²For example, the PROGRESA conditional cash transfer experiment in Mexico used a delayed treatment as control, but future recipients were not informed at the evaluation onset. They were informed only a couple months prior to receiving the program themselves, thus this limits the anticipatory behavior changes (Gertler et al., 2012).

³We partitioned the sample into three age groups: 18-35, 36-50, and 51-65 years old.

asked to formulate a spending plan, and discussed strategies for carrying out their plan successfully with an advisor. At the second planning session, participants received information on opportunities for investing in current or new income-generating activities, savings vehicles, and potential ways to address emergencies. They were then asked to review and revise their original (non-binding) spending plan if they wished to do so.

Expectations of Future Transfer (T3): Individuals assigned to T3 were told that they would receive a grant similar to the one being given to individuals in the T1 and T2 treatments arms, but that they would receive the money in approximately one year. This group received their grant shortly after the endline survey (11 months after treatment), following the same procedures that were used to deliver grants to individuals in T1. They additionally had the option to attend the financial information sessions offered to participants in T2.

Control (C): Individuals assigned to the control group were informed that they would not receive a grant.

All grant recipients received 350,000 Ugandan shillings (UGX; equivalent to \$135 in nominal terms in 2013, or \$337 in 2013 after adjusting for purchasing power parity).⁴ This amount corresponded to two months of the average household income for individuals in the sample. It represents 14.4% of Ugandas annual GDP per capita in 2013 (UGX 2,435,000), similar in magnitude to the UGX 300,000 transfer studied by Blattman et al. (2016).

2.1 Recruitment, Data Collection, and Attrition

We recruited over two thousand participants aged 18 to 60 who were enrolled in one of two TASO clinics located in the rural districts of Masindi in the west and of Soroti in the east.^{5,6} Participants were recruited from TASO clinics, from community drug distribution points run by TASO, and through home visits. A maximum of one TASO client per household was enrolled in the study. Recruitment took place between October 2013 and May 2014. Before completing the baseline survey, participants were informed that some study participants would receive a cash transfer.⁷

⁴The average exchange rate for 2013 was 2584.88 UGX to a US dollar or 1036.87 UGX to a PPP-adjusted US dollar. A 2013 US dollar is equivalent to \$1.38 in 2025.

⁵This age group ranges from the legal age of maturity to the retirement age.

⁶The Masindi center provides care to over 3,800 patients from that district and the surrounding districts of Buliisa, Hoima, Nakasongola, and Kibale. The Soroti center is larger with over 5,900 patients from the Soroti, Kumi, Katakwi, Amuria and Kaberamaido Districts.

⁷At least half of participants have no acquaintances assigned to treatment, and at least three-quarters have no more than one, limiting concerns about spillovers across households.

In total, 2,170 individuals completed the initial baseline survey and were randomly assigned to treatment. We conducted four additional surveys: a short, high-frequency panel comprising three surveys spanning the period from immediately before treatment (i.e. immediately before and after individuals assigned to T1 and T2 received grants) to one month after treatment, plus an endline survey eleven months after treatment. The first high-frequency survey was conducted between one to two weeks before treatment assignments were announced, and the second and third high-frequency surveys were conducted three and six weeks thereafter. The first follow-up survey provides updated pre-treatment data to complement our main baseline, such as scale and scope of income-generating activities immediately prior to grant disbursement. The second and third follow-up surveys allow us to examine the ways that grant recipients made use of the funds they received. Our analysis at midline (endline) focuses on individuals who completed the midline (endline) survey. However, our results are robust to focusing on the sample who completed *all* survey rounds ($N = 1973$, or 90.9% of baseline respondents).

In Table 1, we present summary statistics for the whole sample and at the treatment arm level. The last column tests for and finds no statistically significant differences in baseline characteristics across arms. Each treatment arm contains just under 550 participants. 69 percent of study participants are female, reflecting the composition of TASO's client base.⁸ Approximately half are married, with 8.4% in polygynous unions. The average age among participants is just over 41 years old, and the average level of educational attainment is about 5 years of schooling.

At baseline, most respondents were either working for pay (29.6%) or self-employed (43.3%), with a small fraction doing both (9%). Overall, 85% of people in the sample were either engaged in some form of income-generating activity themselves or had a household member who was engaged in an income-generating activity (IGA). If we also take into account unpaid and domestic work, 97.5% of individuals were economically active. Individuals in our sample were working 19.5 hours a week on average. The vast majority of participants, 81.7%, were in a household owning at least one business. Almost all respondents were involved in the credit market in some way. At baseline, over 96.5% had either some savings or some outstanding loans, and the vast majority had either formal savings or formal loans outstanding (either through banks or through village credit groups). Thus, households were not completely credit-constrained prior to the interventions.

⁸In Africa, women are 2.3 times more likely to contract HIV from men than men are from women from sexual relations, and as much as 60% of people infected by the virus are women (Magadi, 2011).

Table 1: Summary Statistics by Treatment Arm

	FULL SAMPLE	CONTROL	GRANT (T1)	GRANT + PLANNING (T2)	GRANT, DELAYED 1-YEAR (T3)	TEST OF EQUALITY P-VALUE
	(1)	(2)	(3)	(4)	(5)	(6)
Obs. at baseline	2170	548	536	544	542	
Female	69.1%	69.0%	69.2%	69.1%	69.2%	1.00
Married	51.6%	51.1%	52.8%	50.2%	52.2%	0.61
Polygynous	8.4%	8.8%	9.0%	8.1%	7.7%	0.77
Catholic	37.3%	39.2%	37.1%	36.8%	36.0%	0.92
Protestant	45.5%	44.0%	45.0%	46.7%	46.3%	0.83
Age	41.1 (8.6)	41.1 (8.7)	41.1 (8.8)	41.0 (8.7)	41.2 (8.3)	0.59
Education	05.0 (3.8)	05.2 (4.0)	04.8 (3.8)	05.0 (3.8)	05.0 (3.6)	0.69
Working for pay	29.6%	29.0%	28.1%	29.7%	31.7%	0.43
Self employed	43.3%	42.3%	45.4%	43.1%	42.5%	0.60
Hours worked	19.5 (25.1)	19.6 (25.7)	19.7 (25.5)	19.9 (25.3)	18.8 (23.9)	0.67
HH owns business	81.7%	83.8%	80.6%	79.8%	82.7%	0.45
Savings	76.7%	77.7%	74.5%	77.1%	77.3%	0.50
Borrowing	86.5%	87.6%	87.1%	87.2%	84.0%	0.27
Formal savings	15.8%	17.5%	14.8%	15.9%	14.8%	0.86
Formal borrowing	68.7%	67.8%	70.4%	67.0%	69.8%	0.43
Severe food insecure	61.5%	62.3%	63.5%	61.2%	59.0%	0.24
Joint test of prediction, p-value		0.77	0.84	0.96	0.32	

Note: Standard deviations of non-binary outcomes are reported in parentheses. The outcomes “Working for pay”, “self-employed”, and “hours worked” refer to the week prior to the survey. The last column presents the p-values of a F-test of equality across arms. The last row presents the p-values of joint F-tests indicating whether all the listed variables are predictive of the group assignment.

Most respondents (61.5%) were experiencing severe food insecurity at baseline, in spite of their access to savings and credit. Looking at the components of the food insecurity index, we find that 57.3% of the households had no food to eat for at least a full day over the four weeks prior to the baseline survey, and 31.5% of households had no food for three or more days. 49.4% of households had members going to bed hungry for at least a night and 10.5% had members who didn't eat for at least a full day and a full night over that period. Typically, households were eating grains, roots, nuts, and vegetables 4-5 days a week; fruits, fat and oils, and sugars 3-4 days a week; and meat and dairy products only 2 days a week. Together with the fats and oils food group, meat and dairy constitute key food groups for people with HIV/AIDS as they allow them to build up energy reserves necessary to perform daily activities (World Bank, 2007). Consequently, if cash transfers lead participants to eat more of these food groups, then they could see a rise in their productivity.

Attrition is limited and not compounding: 1.7% at midline and 4.7% at endline.⁹ In Table A1 of Appendix A, we regress midline and endline survey completion on the treatment assignment dummies. All point estimates are very small and insignificant at the 5% level. Participants in the *Expectations* group are approximately 2% more likely to complete endline than the control group, but the coefficient is only marginally significant with an unadjusted p-value of 0.06. In Table A2, we regress midline and endline participation on the same demographic characteristics as above and find that they do not jointly explain the dependent variables.

3 Empirical Strategy and Results

To estimate the impacts of the three interventions, we adopt the following ANCOVA specification:

$$y_{i,t} = \beta_0 + \beta_1 T_{1,i} + \beta_2 T_{2,i} + \beta_3 T_{3,i} + \gamma y_{i,t=0} + \delta_{strata} + \varepsilon_{i,t}, \quad (1)$$

where $y_{i,t}$ is the dependent variable for participant i at time $t \in 1 \text{ month}, 12 \text{ months}$; $T_{1,i}$, $T_{2,i}$, and $T_{3,i}$ are indicators for the *Cash*, *Cash with Planning*, and *Expectations* groups, respectively; $y_{i,t=0}$ is the value of dependent variable at baseline, if available; and δ_{strata} are strata fixed effects. In our main results, we report Eicker-Huber-White standard errors and apply the Benjamini-Hochberg (BH) correction to the p-values to further account for multiple hypotheses tests.

⁹24 respondents (1%) completed the second follow-up survey but not the third, and 8 respondents (0.4%) completed the third follow-up survey but not the second. For these respondents, we use the available survey round to construct all midline outcome variables. Unfortunately, 1% of the original participants died during the course of the study. We do not attempt to survey the family members of those who passed away, so they are counted among the attritors.

3.1 Impacts After One Month

3.1.1 Immediate Transfers

In Table 2, we present midline results, constructed from the second and third rounds of the high-frequency panel (surveys conducted one and four weeks after treatment, respectively). At that point, participants in treatment groups T1 and T2 had received transfers, while participants in the expectations group anticipated receiving transfers 11 months in the future. We report impacts on nine main outcomes. We consider four measures of involvement in income-generating activities that capture the impacts of treatment on the economic activities of individuals and households: hours worked by the respondent, which is the sum of paid work hours and hours of own-account work; total business expenditures across all enterprises operated by household members; total business profits; and a count of the number of distinct IGAs that the household was involved in. We capture overall household welfare by measuring food security, dietary diversity, and total household expenditure. Finally, we calculate the total amount saved and the total amount owed by the household. To account for outliers and misreporting, we winsorize the top and bottom 1% of values for these continuous variables.

Results in Table 2 demonstrate that cash grants had large impacts on households' wellbeing and involvement in IGAs, and that the impacts of T1 (*Cash*) and T2 (*Cash with Planning*) were broadly similar. In the month after receiving a transfer, individuals assigned to T1 work approximately 3.6 hours more, relative to the control group, while those assigned to T2 work 6 hours more (BH q-values <0.01 in both cases).¹⁰ Compared to the control group, this represents a 21-37% increase in work hours. Thus, even a large unconditional cash transfer does not discourage work, consistent with (Banerjee et al., 2017). Respondents assigned to the two contemporaneous-transfer groups also have higher business expenditures and business profits in the month after they received their grants. Total business expenditures increase by 78% for the T1 group and by 71% for the T2 group (BH adjusted p-values <0.01). Both groups also see an increase in business profits of 48.3% for T1 (29,000 UGX, BH q-value <0.01) and of 62.6% for T2 (38,000 UGX, BH q-value <0.01). In addition, approximately one in three households in both groups increase the number of IGAs that they are involved in (translating to an average 16.7% increase in T1 and 19.5% increase in T2, BH q-values <0.01 in both cases).

¹⁰The difference between the impacts of T1 and T2 is marginally statistically significant (unadjusted p-value 0.06).

Table 2: OLS Intent to Treat Estimates One Month Post T1/T2 Grants and Eleven Months Prior to T3 Grants

	WORK HOURS	BUSINESS EXPEND.	BUSINESS PROFITS	IGA COUNT	FOOD SECURITY INDEX	DIETARY DIVERSITY INDEX	HH EXPEND.	TOTAL SAVED	TOTAL OWED
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Grant (T1)	3.55*** (0.00) [0.00]	174.36*** (0.00) [0.00]	29.14*** (0.00) [0.00]	0.29*** (0.00) [0.00]	0.27*** (0.00) [0.00]	0.25*** (0.00) [0.00]	99.56*** (0.00) [0.00]	73.05*** (0.00) [0.00]	-26.77 (0.25) [0.28]
Grant + Planning (T2)	6.01*** (0.00) [0.00]	158.36*** (0.00) [0.00]	37.75*** (0.00) [0.00]	0.34*** (0.00) [0.00]	0.27*** (0.00) [0.00]	0.30*** (0.00) [0.00]	70.99*** (0.00) [0.00]	79.73*** (0.00) [0.00]	-40.53* (0.06) [0.08]
Grant, Delayed 1-Year (T3)	2.91** (0.01) [0.02]	-10.05 (0.71) [0.73]	17.10** (0.03) [0.04]	0.08 (0.10) [0.12]	0.08* (0.07) [0.09]	0.11** (0.03) [0.04]	13.05 (0.32) [0.34]	-0.54 (0.97) [0.97]	-37.55* (0.05) [0.06]
Pr(T1 = T2)	(0.06)	(0.65)	(0.39)	(0.30)	(0.92)	(0.33)	(0.06)	(0.64)	(0.52)
Pr(T3 = T1)	(0.60)	(0.00)	(0.19)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.57)
Pr(T3 = T2)	(0.02)	(0.00)	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.86)
R-squared	0.37	0.43	0.25	0.34	0.37	0.23	0.32	0.57	0.59
Control Mean	16.37	223.61	60.28	1.72	0.15	0.18	210.57	131.49	221.92
Control S.D.	22.95	600.18	132.00	0.97	0.93	0.93	262.23	339.32	489.09
Observations	2133	2133	2133	2133	2133	2133	2133	2133	2133

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively, after Benjamini-Hochberg (BH) corrections of the p-values to account for multiple hypotheses tests. We report regression coefficients, the associated Eicker-Huber-White p-values in parentheses, and BH-adjusted p-values in square brackets. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. The Income-Generating Activity (IGA) count captures the number of sources of income of the participant's households. Work hours are the weekly hours worked for wages and in the household's business(es) reported for the week prior to the survey. Total expenditure or household expenditure, business expenditure, business profits, total saved, and total owed are all measured in thousands of UGX. The grants were 350,000 UGX (USD\$135 nominal or USD\$337 in 2013). The average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP. A USD\$ of 2013 was approximately worth 1.38 USD\$ in 2025.

Both T1 and T2 also increased food consumption. We observe a 0.3 standard deviation (SD) increase in food security and dietary diversity in both groups, with BH-adjusted p-values well below 0.01. Participants also eat more of the foods important for productivity, especially for people with HIV/AIDS (World Bank, 2007). Online Appendix Table C3 demonstrates that both T1 and T2 had large positive impacts on consumption of meat and fats – though these treatments also increased consumption of sweets. This increase in protein and fat consumption may play a key role in explaining the increase in work hours of these participants, as we explore further below.

The two contemporaneous-transfer groups also have higher household expenditures after receiving the transfer. Participants in T1 spend 47.3% (approximately 100,000 UGX, BH q-value <0.01) more on household expenditures, while those in T2 increase household spending by 33.7% (71,000 UGX, BH q-value <0.01). Respondents in T1 and T2 also have 55.6% and 60.6% more, respectively, in total household savings relative to the control group following treatment (both BH q-values <0.01). While not significant for T1, the point estimates on the household debt suggest a decrease of 12.1% and 18.3% in the total amount owed by participants in T1 and T2, respectively (BH q-values 0.29 and 0.08, respectively). Overall, participants in T1 and T2 eat more and better, they consume more, work more, invest more in their businesses, make more profit from those businesses, and save more.

Summing impacts on household and business expenditures, we find that participants in T1 increase their spending by approximately 274,000 UGX, which represents 78.3% of the transfer size. Adding these changes in expenditures to impacts on savings and borrowing, we can account for 106% of the amount transferred in during the year and (an oddly exact) 100% of the amount transferred in T2 – though estimates of the change in household spending and assets are measured with error, and the increase in business profits resulting from the transfer may create a multiplier effect.

3.1.2 Expectations of Future Transfers

Next, we focus on the *Expectations* treatment arm (T3) that had yet to receive the transfer at the time of the midline surveys. Though this group had yet to receive any funds, we observe meaningful impacts on several outcomes, typically around 30-50% of the impacts of immediate transfers. In particular, food security and dietary diversity increase by approximately 0.1 SD (BH q-values 0.09 and 0.04, respectively). Similar to participants in T1, people in the *Expectations* group work 2.9 additional hours per week (17.8% increase, BH q-value 0.02) and their business profit increases by 17,000 UGX (28.4%, BH q-value 0.04). Also, similar to the other treatment groups, there is no evidence of dissaving for the *Expectations* group. While savings don't increase, borrowing falls by 16.9% (37,500 UGX, BH q-value 0.06). Unlike the groups who received the grant by this point,

participants in the *Expectations* treatment see little change in household or business expenditures or in their income-generating activities. However, the point estimate on household expenditure is positive with the largest relative increase coming from food expenditure (15% increase), which is also associated with a 14% increase in the number of days meat is consumed (see Online Appendix Table C3, p-value 0.01). This suggests that rising work hours and business profits help households eat more regularly and more diversely, given that net savings go up for this group.

3.2 Impacts After One Year

Endline surveys took place one year after treatment arms T1 and T2 received grants and just before the *Expectations* group received theirs. After one year, groups that had already received the transfer (T1 and T2) still have higher food security (0.17 SD and 0.2 SD, BH q-values 0.08 and 0.03) and dietary diversity (0.4 SD and 0.5 SD, BH q-values <0.01) than the control group. They also have higher household expenditures (BH q-values <0.01), more savings (BH q-values both <0.01), and more durable assets (0.17 SD and 0.1 SD, respectively, BH q-value <0.01 for T1, 0.03 for T2). The mental health of individuals in T1 and T2 is also higher than at baseline (0.17 SD, BH q-values 0.01). Thus, unconditional cash transfers improved households living conditions and wellbeing. Interestingly, though impacts on the total number of IGAs persist, impacts on hours worked are somewhat attenuated by endline: T1 is associated with a 1.43 increase in work hours which is not statistically significant (BH q-value 0.34) while T2 is associated with a marginally significant 2.37 hour increase in hours worked (BH q-value 0.1). Since individuals in the control group work, on average, 15.6 hours per week, we cannot rule out meaningful impacts on hours – but they appear less pronounced after one year than they did one month post-treatment. We see little change in housing conditions for any groups.

By endline, participants in the *Expectations* treatment group who still had yet to receive the grant looked more similar to participants in T1 and T2 than they did 11 months prior. They see a similar increase in food security (0.25 SD, BH q-value 0.01), dietary diversity (0.26 SD, BH q-value 0.01), work hours (2.7 hours, BH q-value 0.05), total household expenditure (23,000 UGX, BH q-value 0.04), and durable assets (0.11 SD, BH q-value 0.03). Once again, there is no evidence of dissaving for this group. As opposed to the other treatment groups however, the *Expectations* group still sees little increase in its income-generating activity count or its mental health.

Table 3: OLS Intent to Treat Estimates One Year Post T1/T2 Grants and Prior to T3 Grants

	WORK HOURS	IGA COUNT	FOOD SECURITY INDEX	DIETARY DIVERSITY INDEX	HH EXPEND.	TOTAL SAVED	TOTAL OWED	HOUSING CONDITIONS INDEX	DURABLE ASSETS INDEX	MENTAL HEALTH INDEX
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Grant (T1)	1.43 (0.26) [0.34]	0.22*** (0.00) [0.00]	0.17* (0.05) [0.08]	0.39*** (0.00) [0.00]	48.37*** (0.00) [0.00]	75.78*** (0.00) [0.00]	27.21 (0.35) [0.42]	0.02 (0.64) [0.69]	0.17*** (0.00) [0.00]	0.17** (0.00) [0.01]
Grant + Planning (T2)	2.37* (0.06) [0.10]	0.20*** (0.00) [0.00]	0.20** (0.02) [0.03]	0.45*** (0.00) [0.00]	35.53*** (0.00) [0.00]	77.48*** (0.00) [0.00]	9.73 (0.72) [0.74]	-0.01 (0.87) [0.87]	0.10** (0.01) [0.03]	0.17** (0.00) [0.01]
Grant, Delayed 1-Year (T3)	2.74* (0.03) [0.05]	0.04 (0.45) [0.50]	0.25*** (0.00) [0.01]	0.26** (0.00) [0.01]	22.80** (0.02) [0.04]	24.07 (0.08) [0.12]	-20.78 (0.40) [0.46]	-0.03 (0.32) [0.41]	0.11** (0.01) [0.02]	0.09 (0.12) [0.16]
Pr(T1 = T2)	(0.47)	(0.72)	(0.71)	(0.50)	(0.28)	(0.93)	(0.54)	(0.55)	(0.11)	(1.00)
Pr(T3 = T1)	(0.31)	(0.00)	(0.31)	(0.16)	(0.02)	(0.00)	(0.07)	(0.16)	(0.23)	(0.19)
Pr(T3 = T2)	(0.78)	(0.00)	(0.50)	(0.03)	(0.23)	(0.00)	(0.21)	(0.46)	(0.75)	(0.20)
R-squared	0.29	0.21	0.13	0.15	0.24	0.30	0.44	0.71	0.35	0.04
Control Mean	15.58	1.71	-1.15	7.78	130.18	96.33	249.38	0.09	-0.12	0.00
Control S.D.	24.04	0.95	1.49	1.54	184.45	231.17	513.38	1.05	0.81	1.00
Observations	2069	2069	2069	2068	2069	2069	2069	2069	2069	2069

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively, after Benjamini-Hochberg (BH) corrections of the p-values to account for multiple hypotheses tests. We report regression coefficients, the associated Eicker-Huber-White p-values in parentheses, and BH-adjusted p-values in square brackets. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions except for the mental health regression since the questions related to that variable were only asked at endline. The Income-Generating Activity (IGA) count captures the number of sources of income of the participant's households. Work hours are the weekly hours worked for wages and in the household's business(es) reported for the week prior to the survey. Total expenditure or household expenditure, business expenditure, total saved, and total owed are all measured in thousands of UGX. The grants were 350,000 UGX (USD\$135 nominal or USD\$337 in 2013). The average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP. A USD\$ of 2013 was approximately worth 1.38 USD\$ in 2025.

3.3 Discussion

Our results suggest that immediate transfers have large, positive impacts on business and household outcomes, and we find no evidence that transfers discourage labor supply. More surprisingly, we find that the promise of future transfers also leads to meaningful immediate changes in food consumption and labor supply, and that these changes translate into longer-term impacts on many household outcomes that are almost as large as the effects of immediate transfers.

While a standard life-cycle model predicts that households expecting a future transfer might borrow to smooth consumption, we do not find any evidence that this mechanism is at play. If anything, households in the expectations treatment appear to pay down their debt instead of borrowing against future income.

Our preferred model, presented in the next section, predicts that individuals who are *certain* of receiving a future transfer should adjust very similarly to those receiving a transfer immediately. The fact that we see large adjustments in the expectations group indicates that large part of the participants in this group believed in our promise. It is entirely possible that some didn't believe our promise or had some doubts as to whether they would actually receive the transfer. The model predicts no adjustment for the former and a smaller adjustment for the latter compared to those who believe that a transfer will come with probability one. Hence, when looking at the average responses in the *Expectations* group, it is not surprising to see smaller adjustments in magnitude than in the groups that received an early transfer.

By endline, the adjustment in the *Expectations* group even slightly exceeds the adjustment in other treatment groups when it comes to food security and work hours. As a result, using a group that has yet to receive a transfer but expects to receive one as a control group can severely bias the effects measured of a contemporaneous transfer. Since the sign of the adjustments for participants in the *Expectations* group is typically the same as that of the other treatment group, but of lesser magnitude, this exercise would lead us to severely underestimate the effect of a contemporaneous cash transfer. For food security and work hours at endline, doing so would even yield negative point estimates and lead to erroneous conclusions. Hence, from a methodological standpoint, the existence of a pure control group can be critical for identification in such settings.

4 Theory

We begin by presenting a simple life-cycle model where infinitely lived individuals maximize their discounted expected utility in the spirit of Hall (1978). We will show that the basic life-cycle model and certain extensions fail to deliver key predictions, especially with regards to the effect of transfers on work hours. Then, we propose an extension of the model supported by the data.

In the model, individuals derive utility from consumption, C , and leisure. Consistent with the summary statistics, we assume that individuals can save and borrow which allows them to transfer wealth across periods. In the basic model, labor income does not depend on labor productivity. For simplicity, labor income is given by work hours times a fixed hourly wage. During each period, individuals earn interest on their savings if net savings are positive or pay interest if they are negative. They work and receive labor income, and they may receive a transfer this period depending on the treatment arm they are in. They choose how much of these current resources to allocate to consumption of a continuous good and to net savings that are carried forward to the next period.

Combining these elements, we obtain the following intertemporal maximization problem where at time t individuals maximize their current and future expected utility by choosing their stream of consumption, work hours, and savings for the current and future periods (denoted by $\{C_t\}$, $\{L_t\}$, and $\{S_{t+1}\}$) :¹¹

$$\begin{aligned} \max_{\{C_t\}, \{L_t\}, \{S_{t+1}\}} E_t[U] &= E_t \left\{ \sum_{s=0}^{\infty} \beta^s [\ln(C_{t+s}) + V(\bar{L} - L_{t+s})] \right\} \\ \text{s.t. } E_t \left\{ \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s [S_{t+s+1} + PC_{t+s} - \underbrace{(1+r)S_{t+s} + W_{t+s}L_{t+s} + T_{t+s}}_{Wealth_{t+s}}] \right\} &= 0 \end{aligned} \quad (2)$$

In the maximization problem, β is the discount factor, $V(\ell)$ is the utility of leisure. Leisure, ℓ , is the difference between available hours $\bar{L}$ and work hours. In particular, we assume that $V(\ell)$ is increasing, but concave in leisure such that $V(\ell) > 0$, $V'(\ell) > 0$, and $V''(\ell) < 0$.¹² For simplicity and tractability, utility is additively separable and depends on log consumption. As a result, the marginal utility of consumption is also decreasing. As we can see from the first order conditions below, this choice entails that the marginal utility of wealth is decreasing in wealth.¹³ S_{t+1} is savings at the end of period t . P is the cost of consumption, which we assume to be constant in time without loss of generality. $r \geq 0$ is the real interest rate on savings that we also assume to be constant, and T is the cash transfer.

From the Lagrangian of the maximization problem, $\mathcal{L}$, we obtain the following First Order Conditions (FOC) for the key choice variables:

¹¹Net savings are chosen at the end of period t , hence the index $t+1$ on that variable. The expectation at time t takes into account the knowledge at this period (and previous periods) only. Hence, $E_t[T_t] = T_t$ for a transfer, T , received this period regardless of whether it is anticipated or a surprise because it is received at period t , regardless. However, $E_t[T_{t+1}] = T_{t+1}$ if an individual anticipates at t that a transfer at $t+1$ will occur. But $E_t[T_{t+1}] = 0$ if the individual does not anticipate receiving a transfer in the future.

¹²As a result, $\frac{\partial}{\partial L} V(\bar{L} - L) = -V'(\bar{L} - L) < 0$, and $\frac{\partial^2}{\partial L^2} V(\bar{L} - L) = -V''(\ell) \frac{\partial \ell}{\partial L} = V''(\bar{L} - L) < 0$.

¹³The concavity of the consumption and leisure part of the utility ensure an interior solution.

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 : \frac{1}{C_t} = P\lambda_t \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial C_{t+1}} = 0 : E_t\left[\frac{1}{C_{t+1}}\right] = \frac{P}{\beta(1+r)} E_t[\lambda_{t+1}] \quad (4)$$

$$\frac{\partial \mathcal{L}}{\partial L_t} = 0 : V'(\cdot) = \lambda_t W_t \quad (5)$$

$$\frac{\partial \mathcal{L}}{\partial S_{t+1}} = 0 : \lambda_t = E_t[\lambda_{t+1}] \quad (6)$$

In the system above, λ_t is the marginal utility of wealth. The Euler equation, Equation (6), states that $\lambda_t = E_t[\lambda_{t+1}]$ and from equations (3) and (4), we obtain $E_t[\frac{1}{C_{t+1}}] = \frac{1}{\beta(1+r)} \frac{1}{C_t}$. Therefore, from the savings FOC, we obtain the smoothing condition common to unconstrained intertemporal models, stating that the expected present value of the marginal utility of consumption and the expected marginal utility of wealth must be equal across periods. It is this desire to smooth consumption and wealth across periods that drives individuals to reoptimize following a wealth shock like a cash transfer.

Next, we investigate the effect of a transfer at time t on the variables of the model. Differentiating Equation (3) with respect to T_t , yields:

$$\frac{\partial}{\partial T_t} \left(\frac{1}{C_t} \right) = P \frac{\partial \lambda_t}{\partial T_t} \leq 0, \quad (7)$$

which is achieved by increasing consumption. Doing the same for the labor FOC yields:

$$\frac{\partial}{\partial T_t} V'(\bar{L} - L_t) = W_t \frac{\partial \lambda_t}{\partial T_t} \leq 0 \quad (8)$$

The equation indicates that the right-hand side of the labor first-order condition declines. For the marginal utility of leisure on the left-hand side to also decrease and restore equality, leisure must increase, given that $V'(\cdot)$ is decreasing in leisure. This implies a reduction in labor supply. As a result, higher consumption following the transfer must be financed by the transfer itself, with part of the transfer saved to smooth consumption over future periods. For an expected future transfer, the basic model predicts that any increase in current consumption must come from dissaving since labor hours fall and the transfer has not been delivered yet. Clearly, this basic structure does not match the empirical findings. Indeed, recall that we find a contemporaneous increase in labor hours and net savings for all treatment groups.

Health is an important consideration in our context, and in many low-income settings. Improvements in health impact utility directly, but they can also increase life expectancy and hence

the discount factor. We consider a case where each period, individuals can invest in their health stock, M_t . We first consider two separate cases where (1) the health stock is an element of the individual's utility, and (2) where the health stock improves the probability of survival in future periods. However, because labor income is not affected by the health stock, the model still predicts a contemporaneous decrease in labor which is again inconsistent with our findings. This is not to say that these standard extensions have no effect on the model's predictions; rather, that extensions leaving the first-order condition of labor unchanged cannot generate an increase in equilibrium work hours.¹⁴

To explain an increase in work hours following a transfer, Banerjee et al. (2015) propose a lumpy durable consumption model where borrowing-constrained individuals borrow and/or dissave and then use those funds together with the transfer to purchase a lumpy durable; this additional capital may raise the marginal productivity of labor, leading to increases in hours worked or individual effort. However, we find that individuals who receive or expect to receive transfers do not buy more expensive durables as a result of treatment, and if anything, point estimates suggest a decrease in borrowing in all treated groups. Therefore, these results allow us to rule out models where the increase in labor supply after a transfer is explained by liquidity constraints and lumpy investment goods.

The key challenge in generating positive labor hour responses in unconstrained models comes from the assumption that leisure is a normal good. Hence, a positive income effect in the form of a cash transfer puts downward pressure on labor supply. Without changing this assumption on leisure or by imposing constraints, we need a channel by which labor demand can increase in response to a positive income shock, which can in turn lead to an increase in equilibrium labor hours.

To do so, we first allow labor income to depend on labor productivity. So, rather than earning a fixed wage per hour worked, this structure could represent more closely piece-rate work arrangements where individuals are paid for their output, or where individuals are self-employed and earning a revenue based on the output of their businesses. The latter idea fits our context well since over 80% of participants' households own at least one business. Moreover, most personal businesses involve agricultural work and animal husbandry requiring physical work and where the revenue depends heavily on the output that can be produced in a set amount of time. Hence, the productivity of labor hours put in these businesses likely matters quite a bit for the revenue they generate. Going back to individuals' health stock, extensive research shows that physical health and

¹⁴If labor income is given by $W_t L_t$ and $U_t = \ln(C_t) + g(M_t) + V(\bar{L} - L_t)$ with $g_M(\cdot) > 0$ and $g_{MM}(\cdot) < 0$. Then, the first order condition for labor is given by $V'(\cdot) = \lambda_t W_t$. Taking the derivative with respect to T_t or T_{t+1} yields a decrease in L_t . The same FOC for labor is obtained if we assume the same functional form for labor income and that survival is endogenous and depends on M . Conditional on surviving to period t , the discounted sum of current and future expected utilities is $E_t\{\ln(C_t) + V(\bar{L} - L_t) + \sum_{s=1}^{\infty} \phi(M_{t+s-1}) \beta^s [\ln(C_{t+s}) + V(\bar{L} - L_{t+s})]\}$. $\phi(M_t)$ is the probability of survival at period $t+1$ with $\phi_M > 0$, $\phi_{MM} < 0$, $\phi(0) = 0$, $\lim_{M \rightarrow \infty} \phi(M) \leq 1$, and $\lim_{M \rightarrow 0} \phi(M) < \infty$.

labor productivity, especially in physically demanding work, are intrinsically linked (e.g. Black et al., 2013). To provide evidence of this, Figure B1 in Appendix B demonstrates that dietary diversity and food security are both positively and strongly correlated with work hours, profit, and profit per work hour in own businesses. Therefore, we believe that in this context, someone's health stock can affect their labor productivity. In particular, we let the labor income take the following shape, $W \cdot f(L, M)$, where $f(\cdot)$ is the product of labor, L is work hours, and M is a productive asset that we think of as physical health, but could represent other variables affecting individual heterogeneity in labor productivity.¹⁵ This way, employment earnings can depend on investment in health.¹⁶ Note that we treat health as a stock variable with the investment in health at time t , m_t , contributing to the stock, M_t , at period t as well such that $M_t = m_t + \delta M_{t-1}$. $0 \leq \delta \leq 1$ is a depreciation factor. We assume that the marginal product of labor and of health are decreasing and we allow for health to be productive. In other words, we assume $f_L(\cdot) > 0$, $f_{LL}(\cdot) < 0$, $f_M(\cdot) \geq 0$, $f_{MM}(\cdot) \leq 0$, $f_{LM}(\cdot) \geq 0$. As mentioned earlier, M could be any asset improving the marginal product of labor. We believe that the stock of health plays a key role here given the type of work done by participants in the study. Moreover, since most participants are self-employed, increases in production inputs other than own labor would yield an increase in the MPL for a broad array of production functions. This is the case for CES production functions where the elasticity of substitution is less than infinity (i.e., if we exclude perfect substitutes, for example). In fact, the results are consistent with both health and own-business inputs being productivity enhancing. For all transfer groups, we see an increase in health investment in the form of better nutrition and for the contemporaneous transfer groups, we also observe an increase in business inputs. In both cases, we also see an increase in work hours, accompanied by an increase in own business profits.

Combining these elements, we obtain the following problem, where m_t is the amount invested in the productive asset and Q is its cost.

$$\begin{aligned} \max_{\{C_t\}, \{L_t\}, \{m_t\}, \{S_{t+1}\}} \quad & E_t[U] = E_t \left\{ \sum_{s=0}^{\infty} \beta^s [\ln(C_{t+s}) + V(\bar{L} - L_{t+s})] \right\} \\ \text{s.t.} \quad & E_t \left\{ \sum_{s=0}^{\infty} \left(\frac{1}{1+r} \right)^s [S_{t+s+1} + PC_{t+s} + Qm_{t+s} - \underbrace{(1+r)S_{t+s} + W_{t+s}f(L_{t+s}, M_{t+s}) + T_{t+s}}_{Wealth_{t+s}}] \right\} \end{aligned} \quad (9)$$

The problem yields the following FOCs:

¹⁵We can also think of the participant's self-employment income as being the output generated by their business times a markup. This formulation for entrepreneurs is similar in spirit to Iacoviello (2005), except that entrepreneurs are both demander and supplier of work here.

¹⁶If $f(L, M) = L$, then we go back to a traditional framework where individuals are working for pay and are paid a wage per hour of work.

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 : \frac{1}{C_t} = P\lambda_t \quad (10)$$

$$\frac{\partial \mathcal{L}}{\partial L_t} = 0 : V'(\cdot) = \lambda_t W_t f_L(\cdot) \quad (11)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial m_t} = 0 : E_t[\lambda_t W_t f_M(\cdot) + \frac{\delta}{1+r} \lambda_{t+1} W_{t+1} f_M(\cdot) + (\frac{\delta}{1+r})^2 \lambda_{t+2} W_{t+2} f_M(\cdot) + \dots - \lambda_t Q] = 0 \\ \lambda_t W_t f_M(\cdot) = \lambda_t Q - \sum_{s=1}^{\infty} (\frac{\delta}{1+r})^s E_t[\lambda_{t+s} W_{t+s} f_M(L_{t+s}, M_{t+s})] \end{aligned} \quad (12)$$

$$\frac{\partial \mathcal{L}}{\partial m_{t+1}} = 0 : E_t[\lambda_{t+1} W_{t+1} f_M(\cdot)] = E_t[\lambda_{t+1}] Q - \sum_{s=2}^{\infty} (\frac{\delta}{1+r})^{s-1} E_t[\lambda_{t+s} W_{t+s} f_M(L_{t+s}, M_{t+s})] \quad (13)$$

$$\frac{\partial \mathcal{L}}{\partial S_{t+1}} = 0 : \lambda_t = E_t[\lambda_{t+1}] \quad (14)$$

Plugging Equation (13) in Equation (12), we obtain $\lambda_t W_t f_M(\cdot) = \lambda_t Q - \frac{\delta}{1+r} E_t[\lambda_{t+1}] Q$. Then, using the savings FOC, we can write

$$f_M(L_t, M_t) = \underbrace{(1 - \frac{\delta}{1+r})}_{0 \leq \rho \leq 1} \frac{Q}{W_t} \quad (15)$$

Next, we investigate the effect of a transfer at time t on the variables of the model. Differentiating Equation (10) with respect to T_t , yields an increase in consumption just like before since the consumption FOC is the same as in the basic model. From Equation (15), we obtain:

$$\underbrace{\frac{f_{ML}(\cdot)}{-f_{MM}(\cdot)}}_{\gamma \geq 0} \frac{\partial L_t}{\partial T_t} = \frac{\partial m_t}{\partial T_t} \quad (16)$$

Equation (16) shows that labor hours and investment in the productive asset comove. Taking the derivative of Equation (11) and plugging in Equation (16), we obtain the following:

$$\left[\underbrace{\lambda_t W_t \gamma f_{LM}(\cdot)}_{\geq 0} + \underbrace{\frac{\text{Increase in the marginal disutility of } \gamma}{\text{additional work hours}} (V''(\cdot) + \lambda_t W_t f_{LL}(\cdot))}_{\leq 0} \right] \frac{\partial L_t}{\partial T_t} = - \underbrace{\frac{\partial \lambda_t}{\partial T_t} W_t f_L(\cdot)}_{\geq 0} \quad (17)$$

The first element in square brackets on the left-hand side captures the positive change in marginal utility for working one additional hour. As mentioned before, when m increases by one unit, L increases by $\gamma \geq 0$ hours. This added investment raises the marginal product of labor.

Hence, $W_t \gamma f_{LM}(\cdot)$ is the added work income stemming from the added investment. Multiplying this quantity by the marginal utility of wealth converts this gain in utils. The second element in the square brackets represents the loss in marginal utility from one additional work hour. $V''(\cdot)$ is the change in the marginal utility of leisure from a reduction in leisure. $\lambda_t W_t f_{LL}$ is the income lost (converted into utils) due to the concavity of the marginal product of labor, relative to the case where the marginal product is constant.

We can see that if the increase in the marginal product of labor from additional investment in a productive asset, like one's physical health, outweighs the added cost of working more, individuals will invest in the asset and will work more. Otherwise, work hours and investment in the asset will fall. Hence, our model can allow for work hours to increase in response to a transfer. To be more precise about the mechanism, it is important to notice that now, not only labor supply can change, but also labor demand. Leisure is a normal good. Hence, the income effect associated with a transfer would lead to a downward shift in the individual's labor supply curve as before. However, if their own work hours become more productive, the labor demand curve increases. As a result, if the labor demand increase outweighs the fall in supply, equilibrium work hours will increase.

Moreover, because of the smoothing condition, we expect the variables above will move the same way following an expected future transfer.¹⁷ From the budget constraint, we can show that if work hours increases, then savings may increase, decrease, or stay the same following a contemporaneous or a future transfer. For example, if the increase in the number of work hours generates a relatively small increase in earnings, an individual may dissave to smooth consumption in prevision of a future transfer. However, if the increase in earnings is large, the person may be able to both smooth consumption and increase savings. If work hours fall, then she will dissave to be able to smooth consumption in prevision of a future transfer and will save to pass wealth in future periods if the transfer is contemporaneous. Table 4 below summarizes the predictions of the model.

What drives reoptimization in the choice variables is the consumers desire to smooth wealth and consumption across periods, which governs how transfers are reallocated intertemporally. In particular, we observe a clear increase in contemporaneous work hours following both a contemporaneous and a future transfer in the data. In the model, this does not reflect a direct effect of smoothing on labor supply; rather, smoothing induces a wealth effect that would by itself reduce work hours, while an increase in labor arises only if the productivity gains from additional investment outweigh the marginal disutility of work. The observed increase in work hours therefore suggests that the productivity channel dominates the pure wealth effect at the relevant margin. We believe this is

¹⁷This is given by Equation (14). Indeed, given that the marginal utilities of wealth must be equal in expectation across periods, a transfer today versus the same transfer in the future will have the same effect on wealth if this future transfer is expected with certainty. As mentioned before, if there is some uncertainty around whether the future transfer will actually be received, adjustments will be smaller in magnitude, compared to an immediate transfer.

plausible given that 7 out of 10 participants are working (for pay or self-employed) and that they work only 20 hours per week on average at baseline. We consider alternative channels below.

Table 4: Model Predictions

Gains in MPL > loss from extra work		Gains in MPL < loss from extra work		$f(L, M) = L$ Hall(1978)	
$T_t \uparrow$	$T_{t+1} \uparrow$	$T_t \uparrow$	$T_{t+1} \uparrow$	$T_t \uparrow$	$T_{t+1} \uparrow$
$C_t \uparrow$	$C_t \uparrow$	$C_t \uparrow$	$C_t \uparrow$	$C_t \uparrow$	$C_t \uparrow$
$L_t \uparrow$	$L_t \uparrow$	$L_t \downarrow$	$L_t \downarrow$	$L_t \downarrow$	$L_t \downarrow$
$M_t \uparrow$	$M_t \uparrow$	$M_t \downarrow$	$M_t \downarrow$	$M_t = 0$	$M_t = 0$
$S_{t+1} \uparrow \downarrow$	$S_{t+1} \uparrow \downarrow$	$S_{t+1} \uparrow$	$S_{t+1} \downarrow$	$S_{t+1} \uparrow$	$S_{t+1} \downarrow$

Note: The direction of arrows indicate the predicted change in the variables following a transfer in the current period or a transfer in a future period. An upward (downward) arrow indicates an increase (decrease). An upward and a downward arrow together indicate an indeterminate sign for the change in the variable. Finally, equality to 0 indicates that the variable remains equal to 0.

4.1 Psychological Channels

One could think that the driving force behind the decision to reoptimize may not only be the individuals' desire to smooth consumption and wealth, but also a psychological effect increasing productivity and stemming from better prospects about the future (Banerjee et al., 2025). Indeed, if the marginal product of labor increases due to a psychological effect, then we may observe similar movements as the model above would predict if the gains from investing in the asset and working more outweighs the cost of doing so. In particular, if we consider M_t as the *total* health of individuals, we could imagine mental health, $h(\cdot)$, to be a component of this variable. If current and anticipated future transfers lead individuals to be more hopeful about the future and better prospects affect mental health, then we could imagine that mental health may depend on the expectation of transfers. In particular, we could assume the following:

$$h(X_t) = h(T_t + (\frac{1}{1+r})T_{t+1} + (\frac{1}{1+r})^2T_{t+2} + \dots),$$

and

$$M_t = m_t + \delta M_{t-1} + h(\cdot),$$

where $h(X)$ is an increasing function of the discounted sum of current and anticipated future

transfers with $h_X(\cdot) > 0$. In this case, Equation (16) becomes:¹⁸

$$\frac{\overbrace{f_{ML}(\cdot)}^{\gamma \geq 0}}{-f_{MM}(\cdot)} \frac{\partial L_t}{\partial T_t} \underbrace{-h_X(X_t)}_{\xi \leq 0} \frac{\partial X_t}{\partial T_t} = \frac{\partial m_t}{\partial T_t} \quad (18)$$

The psychological channel changes the mapping from T_t to m_t (physical investment). However, we can show that Equation (17) determining the sign of $\frac{\partial L_t}{\partial T_t}$ remains the same as before. What differs is that the product of labor is increasing in both investment in physical health and from better prospects about the future. Hence, an increase in work hours here may or may not be accompanied by an increase in investment in physical health. Indeed, Equation (18) indicates that an increase in work hours may be accompanied by a decrease in physical health investment if the psychological effect of the transfer is sufficiently large.

While we certainly can't rule out the presence of a psychological effect of the transfers given that we observe an increase in health investment, we neither have strong evidence for it. We see an increase in an index of mental health for the treatment groups that received a contemporaneous transfer, but not a significant increase for the treatment group that is promised a transfer in the future. However, as we present below, one of the components of the index measures participants' prospect about their life five years ahead. There, we observe a 0.20 SD (or 8.5%) increase for T1 and T2. There is a smaller increase of 0.12 SD (or 5.1%) for individuals in the *Expectations* treatment that is marginally significant (non-adjusted p-value 0.05). On the other hand, people in this latter group do not appear to be more hopeful about their current state as the breakdown of our main effects show below.

5 Potential Mechanisms

To take into account the number of hypotheses to test, we aggregated multiple variables into the indexes analyzed above. To better understand the movements observed, we also look at the variables that compose the indexes and focus on the changes observed at endline unless stated otherwise.

Food Security: This index captures the extent to which household members went to bed hungry, did not eat for complete days, and did not have any food in the house during the 4 weeks prior to the survey. At midline, all treated individuals see a 13.4-32.8% decline in the number of days there is no food at home (compared to the control group). T1 and T2 experience an approximate 35% drop in the number of days they went to bed hungry (see Table C1). By endline, all groups continue

¹⁸Note that if we assume that only a psychological effect affects the product of labor and there are no productive assets, then the product of labor becomes $f(L_t, h(T_t + \frac{1}{1+r}T_{t+1} + (\frac{1}{1+r})^2T_{t+2} + \dots))$. In a static one-period model $f(\cdot) = f(L_t, h(T_t))$, consistent with Banerjee et al. (2025).

to report a 9.8-21.5% drop in the number of days without food at home and 15.7-24.5% decline in the number of days that they went to bed hungry (see Table C2). This suggests that they eat more and more often.

Dietary diversity: This variable captures whether different foods were consumed in the 7 days prior to the survey such as cereal, nuts, vegetables, meat, etc. To understand where the movements are coming from, we look at the number of days over the week prior to the survey different food items were consumed. By endline, participants in the treatment groups see a 14-26.3% increase in the number of days they consume meat, 37.4-46.7% increase in days eating dairy products, and, for T1 and T2, 10.6-12.9% in their consumption of fats and/or oils, compared to the control group (see Table C4). These particular food groups are essential for people with HIV to build up energy reserves to complete their daily activities (World Bank, 2007). Not only do treated groups eat more and more often, but they eat more of the food groups that can improve their energy reserves, consistent with an increase in productivity from better health.

Business expenditures: Health may be an important driver of productivity, but so are business inputs (since the majority of participants have their own businesses).¹⁹ The broad category of business expenditures captures the amount spent on the business in the past three weeks such as rent, machines and tools, other inputs, transport, wages, and inventories. We find little change for treated participants in the *Expectations* group.²⁰ Participants who received the transfer early spend 10,000-12,500 UGX (55-68%) more on rent, 4,000-5,800 UGX (56.7-81.3%) on machines and tools, 22,400-31,700 UGX (64-90.5%) on other inputs, 7,400-11,400 UGX (35.8-55.2%) on employee wages, and mostly on added inventories with the intention of reselling where expenditures increase by 90,400-98,300 UGX (79.5-86.3%) compared to the control group. When focusing on the changes in spending, it appears that participants in T1 and T2 build up their inventories of the goods they resell, they rent larger and/or better spaces, and increase their non-capital inputs as well as the number of and/or the skill level of their employees as reflected by higher expenditures on wages and salaries. At least some of these investments are likely to improve the productivity of their businesses. Consistent with this, we find that revenues generated by the participants' businesses increase by 67% and 64% for T1 and T2, respectively (142,800 UGX and 135,200 UGX). While not significant, the point estimate on revenues is also positive for participants in T3 (17%, a 36,100 UGX increase). From Table 2, we also found that business profits went up in all treatment groups by 48%, 63%, and by 28% for T1, T2, and T3, respectively.

¹⁹Note that the business expenditure module was only asked after one month.

²⁰We see some indication of an increase on transport expenditures which captures spending on transport of products (final and intermediary), of employees, and for their own movements (p-value 0.04, see Table C5). For that category, point estimates for the three treatment groups range between 3,400-7,000 UGX, relative to the control group with the smallest increase for the *Expectations* group. However, this category represents only about 8% of all business expenditures at baseline and falls to 5% at midline.

Income-Generating Activities: To complement the previous analysis, we look at the different sources of profit the participants have at endline. Participants in T1 and T2 are 9% and 12% more likely to raise animals or grow crops for profit, and their number of other businesses increases by 21% and 12% after treatment, compared to the control group (see Table C7). We find no statistically significant effect for the *Expectations* treatment arm.

Work hours: The variable measures the number of hours spent in paid employment and in self-employed work in the seven days before the survey. We find that the increase in total work hours for the treatment groups is driven by a rise in self-employment hours (see Table C9). Self-employment represents over 60% of the participants' work hours in all groups at baseline. Compared to the control group, treated individuals work 25-69% more on their own businesses after one month, and 17-33% more after one year.²¹ This result is consistent with the modeling assumptions. In particular, we assumed that the labor income was the product of a wage and the *product of labor* rather than a wage times the number of hours worked. The latter corresponds more closely to employment paid per hour where the labor income only increases in the number of hours worked, but not in how productive those hours are. We saw that this modeling assumption would predict a fall in work hours and no change in productive assets following a contemporaneous or the promise of a future transfer. In fact, the point estimate on the number of hours worked for a wage is negative for T1 and T2. Allowing labor income to depend on the work output represents more closely self-employment where the self-employed can earn more by producing more and not necessarily and solely by working more. Hence, if there is a way to increase productivity by way of a productive asset, a contemporaneous or the promise of a future transfer can lead self-employed individuals to invest in that asset to increase their productivity and work more if the rise in labor income from added production exceeds the disutility of working more which is in line with our findings.

Taken together, these results suggest that the self-employed work hours of the treated participants do appear to be more productive. Participants in T1 and T2 eat more often and eat foods important to improve one's energy reserves. They grow their businesses and work more in these businesses and while they spend more on building the businesses, they experience a larger increase in business revenues leading them to increase their profits. Participants in the *Expectations* group do not see any large changes that would suggest a growth in the scale of their businesses. Yet, they work more in their businesses and also generate more profits. However, they do eat more and better foods like the other treated participants, suggesting that better health also raises the productivity of self-employed hours.

Mental Health: As mentioned in subsection 4.1, it is possible that receiving or expecting to receive cash transfers also encourages individuals to work more. The main results suggest an

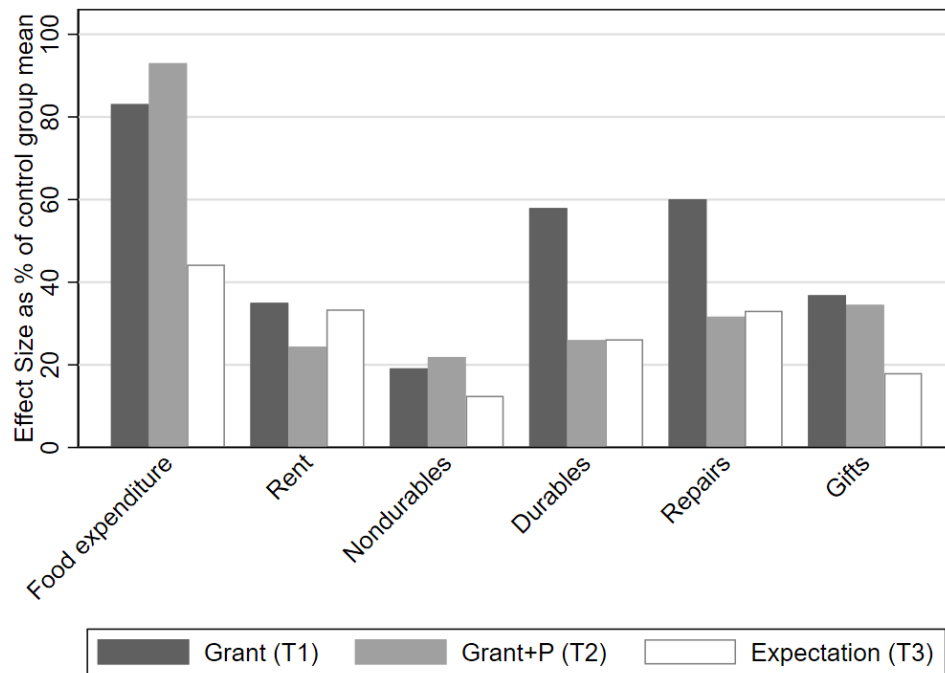
²¹There is a decrease in hours worked for wage and spent searching for work for T1 at midline, but not at endline.

improvement in mental health for participants that received the transfer, but not necessarily for participants in the *Expectations* group. The mental health index is the combination of four variables: a measure of self-esteem, a measure of the participants' locus of control, capturing the beliefs in their own ability to control what happens in their lives, measures of perception of current and of future well-being. No clear patterns emerge for the first two variables. The point estimates are negative and very small for T1, while they are positive and larger (but still small) for T2, and of opposite signs for T3 (see Table C10). There is more to be said for the variables measuring the perception of well-being. The point estimate for the perception of well-being at the time of the survey is positive for all treated groups with an increase 0.18 SD for T1, and smaller and statistically insignificant for T2, and T3. Yet, the perceived well-being five years ahead increases for all treated groups by approximately 0.2 SD (or approximately 8%) for T1 and T2 (unadjusted p-value <0.01, and 0.12 SD (or 5%) for T3, but marginally significant (unadjusted p-value 0.05).²² Hence, there is some support for the hypothesis that receiving or expecting a transfer can impact certain aspects of mental health and that this increase can lead individuals to work more. However, the evidence is only suggestive given the mixed results on self-esteem and locus of control. It is also important to reiterate that unlike other outcomes, the mental health variables were only measured at endline. Hence, we do not control for baseline mental health which could differ between groups.

Household expenditures: Next, we break down the total household expenditure variable into its components. We find that participants in treatment groups see a significant increase in food, nondurables, rent, durables (see next paragraph), dwelling, dwelling repairs (for T1), and gift expenditures (for T1 and T2), as shown in Figure 1 (see Table C12). Nondurables essentially capture goods that are not permanent other than foods such as expenditure on cleaning products, toilet paper, cosmetics, etc. The effect sizes range from 18% to 24%. We see no change on clothing or schooling expenditures, but the point estimates are positive. Consistent with the improvements in food security and dietary diversity, we observe large relative increase in food expenditure of 83% for T1, 93% for T2, and 44% for T3 (3,800, 4,300, and 2,000 UGX). Participants also report spending more on rent (and to repair their dwellings for T1) suggesting an improvement in certain aspects of their living conditions as we investigate next. Participants in T1 and T2 give out more money to religious institutions, charity, family and friends. Taken together, the results point to a clear increase in spending and consumption for individuals in the treated groups.

²²The percentage changes are calculated on the raw scores (not reported).

Figure 1: Household Expenditure Effect Sizes in Percent of the Control Mean After Twelve Months



Housing Conditions and Durables: In the main results above, we found that rent increased for all treatment groups. However, as Table C13 shows we find no clear changes in the number of rooms per household member or in the dwelling materials we measured. It is entirely possible that the improvement was made on other aspects of the dwelling since we measured only a few aspects.

We also saw that durable assets increased for all treated groups. The results on household expenditures indicate that most of the purchases of these goods take place by midline for T1 and T2, while it appears more spread out for T3. We further investigate which durable goods the treated participants and their households purchased. 75% of households own one or more cellphones at baseline. Seven out of ten households own at least one radio. In the treatment groups, we find that approximately one in ten households purchase an additional cellphone (12%). About the same number of households purchase a(n additional) radio (10%). For T2 and T3, we observe little other evidence of meaningful changes in other durables such as cars, motorcycles, bicycles, or in dwelling material. This suggests that for the treatment groups, the increase in durable spending is largely coming from phone and radio purchases rather than pricier durable purchases. As we saw in the main results, participants in T1 tend to be less frugal compared to participants in T2 who received financial planning sessions. For the former group, 7% of participants and their households purchased a television, with one household purchasing a car. While a few households that had members participating in the first treatment arms purchase more expensive durables, it is not a

general observation among treated individuals. The reason that we point this out is that models of lumpy durables purchases can also generate increases in work hours. For example, Banerjee et al. (2015) suggest a model where households want to purchase an expensive durable, but they are credit constrained and unable to do so even by working more. However, following a large transfer, they can borrow what they can, driving down their net savings, use the full amount of the transfer, and work more, and then buy the expensive durable good. As mentioned above this doesn't seem to emulate the general movements we observe given that durable purchase increases are centered around cheaper durables like phones and radios, and that households increase their net savings.

Saving and Borrowing: We find that households with a member participating in any treatment group see a large increase in informal savings and that participants that received a transfer by endline also have more formal savings (see Table C16). Households with a member in T1 experience a two-fold increase in informal savings (99% increase or 24,200 UGX), save 61% more in banks (12,000 UGX) and 49% more in village savings accounts (15,700 UGX). For T2, we observe a 128% increase (31,300 UGX) in informal savings, a 59% increase in bank savings (11,600 UGX), and a 78% rise in village savings accounts (24,800 UGX). In the *Expectations* group, we find an increase in informal savings of 42% (10,200 UGX). There is no significant increase in other savings devices, but the point estimates are positive. There is little change in savings in cooperative savings institutions. The patterns are similar one month after treatment assignment for savings. There are some nuances when it comes to borrowing patterns that vary over time (see Tables C17 and C18). One month after participants in T1 and T2 receive the transfer, we find a negative point estimate on the total amount borrowed in all treatment groups (statistically significant for T2 and T3). When looking at the amount borrowed by sources ranging from family members, banks, NGOs, to employers and shopkeepers, we find that no individual source drives the negative point estimate on the aggregate for T2 and T3. However, the point estimate on most sources of debt and borrowing are negative suggesting a general reduction in debt for this group. This is also true for participants in T1 who also see a significant decrease in the amount they owe in school fees. Given that savings increase, the results indicate an increase in net savings. By endline, participants in T1 and T2 received their transfer 12 months prior. The amount saved by households with a member participating in these groups is still much higher than it was before the experiments by roughly 75,800 and 77,500 UGX, respectively. This time however, the point estimate on the total amount borrowed becomes positive, but remains insignificant, and net savings are still increasing. When looking at the different sources of debt and borrowing, we find meaningful and significant increases of in-kind borrowing for the contemporaneous transfer groups at endline. Households with a participant in T1 also borrow more from family members, microfinance organizations, and NGOs. At the same time, however, the point estimates remain negative for many of the other sources of borrowing and debt for these groups. In the *Expectations* group, we see some evidence of an

increase in borrowing from family members and in-kind, with a decrease in school fees owed and in wage advances (p-values above 0.07). Overall, the results indicate that total savings are increasing and total debt decreasing by endline in all treatment groups.

6 Conclusion

Almost all interventions intended to reduce poverty have the potential to give people hope. Few impact evaluations explore the extent to which interventions work by changing aspirations, psychological wellbeing, and similar determinants of individual heterogeneity in labor productivity – rather than through more conventional economic channels such as borrowing, information, or transaction costs constraints. We compare the promise of transferring cash in a year to transferring cash immediately (and we compare both to no cash transfer). We argue that this provides insight into why the labor response to liquidity is positive, specifically by separating out shifting expectations from the direct liquidity effect of receiving cash.

Individuals promised future money start or expand income-generating activities in anticipation of the later, further infusion of capital. Participants that received an immediate transfer increased household consumption spending and work hours, improved their health through better food and more frequent food consumption, invested more in household businesses, and increased net savings. This is true both one month after and 12 months after receiving the transfer. They also saw an increase in the profits generated by their businesses which was measured after one month. Participants who were promised a future transfer also improve their health through food consumption and increase their work hours, leading to an immediate increase in business profits, and by endline, overall household consumption. One month after treatment, the increases in dietary diversity, food security, and consumption spending observed in the *Expectations* treatment arm are about 30-50% the size of the increase among participants that received the immediate transfer. By endline, the adjustment in work hours is roughly the same in all treatment groups. This suggests that much of the observed impact of cash is, in fact, the effect of changing expectations about the future – and the improvements in the physical and psychological determinants of individual labor productivity – leading to higher investment in income-generating activities by those in extreme poverty. This does not imply that the cash does not have a direct effect that operates by relaxing liquidity constraints, but that represents only one part of the direct effect of a cash transfer.

The results serve as a methodological warning to rolled-in evaluation designs: promising one group future services or transfers shifts their behavior in potentially important directions. In this case, ignoring such effects would have led to a large underestimate of the treatment effect of receiving cash.

Separating shifting expectations from relaxation of other constraints could be fruitful in several other domains. For example, many education interventions improve learning outcomes for children, and also lead to increased investment in time and effort by households and the children in their education due to their changed perception in the return to such investment. Exactly for this reason, many education papers track both learning outcomes and household investments in education for their children. Without having an estimate for the impact of the improved investment level from the household, the returns to the education intervention may be misconstrued to be to the actual resources of the intervention rather than to the accompanying investment by household. A similar issue applies to studies that deliver cash to entrepreneurs on the pretense of then studying the returns to capital. With capital, the expected return to labor changes, and likewise labor inputs shift (this confound has been discussed in many papers, for instance in de Mel et al. (2008)).

The fact that expectations matter is no surprise. Yet interpretation of treatment effects of many interventions often focuses more on the resource aspect of the intervention, treating that as the primary driver and attributing the causal process to models focused on those resources. Here we learn that in the case of a lump-sum cash transfer intervention in Uganda, the shift in expectations was economically important, creating a direct treatment effect comparable to the impact of the cash itself.

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ONLINE APPENDIX

A Attrition

Table A1: Attrition by Treatment Arm

	COMPLETED MIDLINE (1)	COMPLETED ENDLINE (2)
Grant (T1)	0.01 (0.43)	0.01 (0.64)
Grant + Planning (T2)	0.00 (0.90)	0.00 (0.82)
Grant, Delayed 1-Year (T3)	0.01 (0.41)	0.02 (0.06)
Observations	2170	2170

Note: We report coefficient regressions and p-values in parentheses for heteroskedasticity-robust standard errors. We regress whether individuals completed the midline or endline survey in Column (1) and Column (2), respectively, on dummies indicating the treatment arm they were assigned to. We include Strata fixed effects in both regressions.

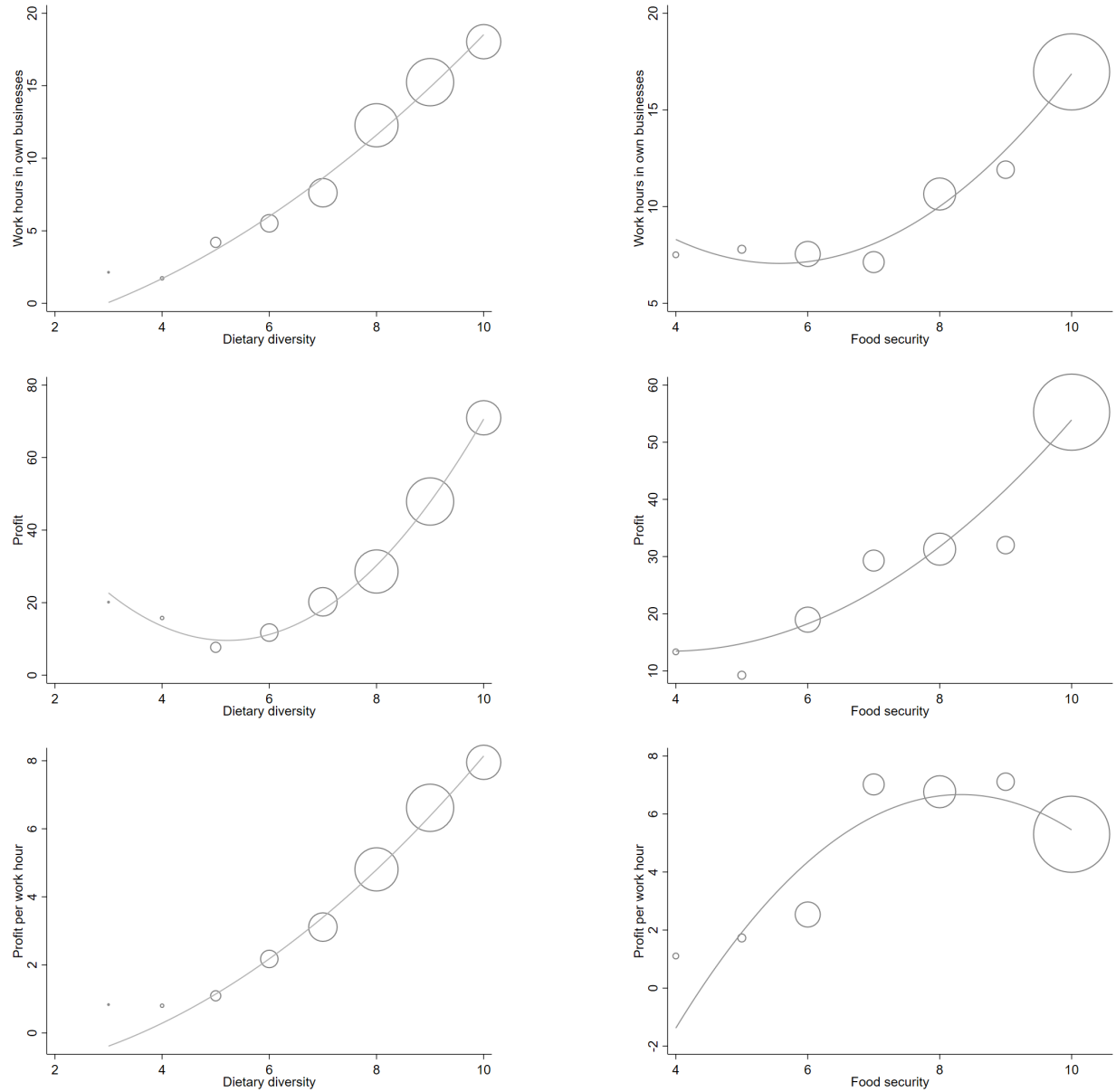
Table A2: Attrition and Demographics

	COMPLETED MIDLINE (1)		COMPLETED ENDLINE (2)	
Female	0.04	(0.33)	0.07	(0.17)
Married	0.00	(0.82)	0.01	(0.45)
Polygynous	0.01	(0.15)	0.01	(0.46)
Catholic	0.01	(0.52)	0.01	(0.63)
Protestant	0.01	(0.41)	0.00	(0.77)
Age	0.00	(0.30)	0.00	(0.16)
Education	-0.00	(0.27)	0.00	(0.86)
Working for pay	0.01	(0.01)	0.01	(0.47)
Self employed	0.00	(0.46)	0.01	(0.36)
Hours worked	-0.00	(0.65)	0.00	(0.48)
HH owns business	0.00	(0.62)	0.02	(0.13)
Savings	0.00	(0.65)	0.01	(0.31)
Borrowing	-0.00	(0.62)	0.02	(0.25)
Formal savings	-0.00	(0.80)	-0.00	(0.90)
Formal borrowing	-0.01	(0.19)	-0.00	(0.66)
Severe food insecure	0.01	(0.09)	0.00	(0.97)
Joint test of prediction, p-value	0.22		0.13	
Observations	2071		2071	

Note: We report coefficient regressions and p-values in parentheses heteroskedasticity-robust standard errors. We regress whether individuals completed the midline or endline survey in Column (1) and Column (2), respectively, on demographics and Strata fixed effects. The last row presents the p-values of joint F-tests of significance for the demographics listed in the table.

B Correlations with Food Security and Dietary Diversity

Figure B1: Correlations of Work Hours, Profit, Profit per Work Hour in Own-Businesses with Dietary Diversity and Food Security



Note: We plot, in order, the average work hours, profit and profit per work hours in own businesses by values of the dietary diversity (left figures) and food security indexes (right figures) at baseline. The size of the circles is proportional to the number of individuals at those value of the indexes. We also include a quadratic fit based on the entire baseline data, rather than just the plotted averages.

C Additional Tables

Table C1: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Food Security

VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Food Security	0.27 (0.00)	0.27 (0.00)	0.08 (0.07)	0.15 (0.93)	(0.92)	(0.00)	(0.00)	2133
Panel B: Components								
No food at home	-0.21 (0.00)	-0.22 (0.00)	-0.09 (0.03)	0.67 (0.83)	(0.75)	(0.00)	(0.00)	2132
Went to sleep hungry	-0.23 (0.00)	-0.21 (0.00)	-0.03 (0.44)	0.65 (0.84)	(0.60)	(0.00)	(0.00)	2132
Did not eat for a day	-0.03 (0.12)	-0.04 (0.02)	-0.02 (0.23)	0.08 (0.34)	(0.49)	(0.63)	(0.19)	2133

Note: Panel A reproduces the results of Table 2. Eicker-Huber-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are: (1) How often there was no food at home in the 4 weeks prior to the survey; (2) How often members of the households went to bed hungry; (3) How often members of the households did not eat for 24 hours. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C2: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Food Security

	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	N
STATISTICS VARIABLES	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Food Security	0.17 (0.05)	0.20 (0.02)	0.25 (0.00)	-1.15 (1.49)	(0.71)	(0.31)	(0.50)	2069
Panel B: Components								
No food at home	-0.05 (0.24)	-0.07 (0.10)	-0.11 (0.02)	0.51 (0.78)	(0.65)	(0.23)	(0.44)	2067
Went to sleep hungry	-0.09 (0.04)	-0.11 (0.01)	-0.14 (0.00)	0.57 (0.78)	(0.61)	(0.23)	(0.50)	2065
Did not eat for a day	-0.02 (0.32)	-0.01 (0.38)	-0.01 (0.65)	0.07 (0.29)	(0.90)	(0.53)	(0.62)	2067

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are: (1) How often there was no food at home in the 4 weeks prior to the survey; (2) How often members of the households went to bed hungry; (3) How often members of the households did not eat for 24 hours. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C3: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Dietary Diversity

		GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			
		(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	N
STATISTICS	VARIABLES	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results									
Dietary Diversity		0.25 (0.00)	0.30 (0.00)	0.11 (0.03)	0.18 (0.93)	(0.33)	(0.00)	(0.00)	2133
Panel B: Components									
Days household ate cereals		0.22 (0.17)	0.33 (0.04)	0.04 (0.81)	4.64 (2.84)	(0.49)	(0.25)	(0.07)	2133
Days household ate tubers		-0.06 (0.63)	-0.05 (0.70)	0.11 (0.38)	5.63 (2.20)	(0.91)	(0.19)	(0.21)	2133
Days household ate nuts		0.13 (0.27)	0.08 (0.47)	-0.10 (0.36)	4.47 (2.37)	(0.66)	(0.04)	(0.09)	2133
Days household ate vegetables		-0.13 (0.23)	-0.14 (0.18)	-0.09 (0.39)	5.76 (1.90)	(0.95)	(0.69)	(0.63)	2133
Days household ate meat		0.59 (0.00)	0.60 (0.00)	0.27 (0.01)	1.99 (1.79)	(0.91)	(0.00)	(0.00)	2133
Days household ate fruit		0.10 (0.39)	0.08 (0.49)	0.22 (0.08)	4.96 (2.60)	(0.88)	(0.34)	(0.28)	2133
Days household ate dairy		0.28 (0.04)	0.27 (0.05)	0.11 (0.44)	1.74 (2.61)	(0.95)	(0.20)	(0.23)	2133
Days household ate fats or oils		0.45 (0.00)	0.43 (0.00)	0.20 (0.13)	3.57 (2.63)	(0.86)	(0.05)	(0.08)	2133
Days household ate sweets		0.36 (0.01)	0.51 (0.00)	-0.05 (0.74)	4.08 (3.00)	(0.30)	(0.01)	(0.00)	2133
Days household ate spices		-0.04 (0.42)	-0.09 (0.10)	-0.09 (0.12)	6.87 (0.85)	(0.41)	(0.47)	(0.91)	2133
Days household consumed soda		0.11 (0.10)	0.12 (0.06)	0.07 (0.29)	0.38 (1.05)	(0.83)	(0.57)	(0.44)	2133
Days household consumed alcohol		0.03 (0.63)	0.05 (0.27)	-0.04 (0.28)	0.13 (0.76)	(0.62)	(0.16)	(0.04)	2133

Note: Panel A reproduces the results of Table 2. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the number of days during the 7 days prior to the survey that certain food groups were consumed. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C4: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Dietary Diversity

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Dietary Diversity	0.39 (0.00)	0.45 (0.00)	0.26 (0.00)	7.78 (1.54)	(0.50)	(0.16)	(0.03)	2068
Panel B: Components								
Days household ate cereals	0.13 (0.43)	0.18 (0.27)	-0.04 (0.80)	3.26 (2.90)	(0.76)	(0.29)	(0.17)	2068
Days household ate tubers	-0.00 (1.00)	0.06 (0.60)	0.04 (0.75)	5.84 (1.95)	(0.61)	(0.76)	(0.84)	2068
Days household ate nuts	-0.31 (0.01)	-0.24 (0.04)	-0.16 (0.16)	5.13 (2.10)	(0.53)	(0.21)	(0.53)	2068
Days household ate vegetables	0.28 (0.03)	0.26 (0.05)	0.14 (0.28)	4.93 (2.45)	(0.88)	(0.28)	(0.36)	2067
Days household ate meat	0.47 (0.00)	0.47 (0.00)	0.25 (0.02)	1.79 (1.78)	(0.99)	(0.05)	(0.04)	2066
Days household ate fruit	0.28 (0.08)	0.16 (0.33)	0.19 (0.24)	3.55 (3.00)	(0.43)	(0.53)	(0.85)	2066
Days household ate dairy	0.48 (0.00)	0.50 (0.00)	0.40 (0.00)	1.07 (2.23)	(0.92)	(0.60)	(0.53)	2066
Days household ate fats or oils	0.38 (0.01)	0.46 (0.00)	0.25 (0.08)	3.57 (2.69)	(0.59)	(0.37)	(0.16)	2066
Days household ate sweets	0.65 (0.00)	0.54 (0.00)	0.32 (0.04)	3.96 (2.95)	(0.49)	(0.03)	(0.14)	2066
Days household ate spices	-0.10 (0.24)	-0.06 (0.42)	-0.07 (0.40)	6.75 (1.22)	(0.68)	(0.72)	(0.97)	2068
Days household consumed soda	0.24 (0.00)	0.09 (0.10)	0.08 (0.16)	0.24 (0.78)	(0.02)	(0.02)	(0.85)	2066
Days household consumed alcohol	0.03 (0.46)	0.00 (0.93)	0.11 (0.03)	0.10 (0.64)	(0.51)	(0.17)	(0.03)	2064

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the number of days during the 7 days prior to the survey that certain food groups were consumed. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C5: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Business Expenditures

VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Business Expenditure	174.36 (0.00)	158.36 (0.00)	-10.05 (0.71)	223.61 (600.18)	(0.65)	(0.00)	(0.00)	2133
Panel B: Components								
Business revenue	142.84 (0.00)	135.15 (0.00)	36.10 (0.22)	211.78 (586.57)	(0.85)	(0.01)	(0.00)	2133
Business rent expenditures	12.51 (0.00)	10.14 (0.00)	3.11 (0.32)	18.41 (52.82)	(0.55)	(0.01)	(0.04)	2133
Inputs expenditures	31.70 (0.00)	22.44 (0.00)	3.73 (0.54)	35.04 (100.93)	(0.24)	(0.00)	(0.00)	2133
Equip expenditures	5.79 (0.01)	4.04 (0.04)	0.00 (1.00)	7.12 (30.49)	(0.43)	(0.00)	(0.04)	2133
Repairs expenditures	1.15 (0.03)	0.37 (0.47)	0.14 (0.77)	2.06 (8.16)	(0.17)	(0.06)	(0.64)	2133
Transport expenditures	7.06 (0.00)	5.80 (0.00)	3.41 (0.04)	7.78 (28.43)	(0.53)	(0.07)	(0.22)	2133
Wages expenditures	11.37 (0.00)	7.37 (0.03)	3.65 (0.26)	20.60 (56.53)	(0.27)	(0.02)	(0.27)	2133
Stock expenditures	90.43 (0.00)	98.25 (0.00)	-23.25 (0.20)	113.82 (399.02)	(0.75)	(0.00)	(0.00)	2133
Other expenditures	4.28 (0.03)	1.76 (0.29)	-0.93 (0.57)	11.69 (34.58)	(0.19)	(0.01)	(0.10)	2133

Note: Panel A reproduces the results of Table 2. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the amount spent in thousands of UGX over the 3 weeks prior to the survey on different goods and services. Each row of the second panel represents a regression on a different outcome. We report regression coefficients and the p-values in parentheses associated with the Eicker-White standard errors in columns T1, T2, and T3. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C6: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Income-Generating Activities

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
IGA Count	0.29 (0.00)	0.34 (0.00)	0.08 (0.10)	1.72 (0.97)	(0.30)	(0.00)	(0.00)	2133
Panel B: Components								
Cashcrops and animal husbandry	0.13 (0.00)	0.11 (0.00)	0.03 (0.36)	1.23 (0.80)	(0.55)	(0.01)	(0.06)	2133
Number of other businesses	0.16 (0.00)	0.23 (0.00)	0.05 (0.08)	0.49 (0.59)	(0.04)	(0.00)	(0.00)	2133

Note: Panel A reproduces the results of Table 2. Eicker-Huber-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are: (1) Whether the participant's household grows crops and/or animals for profits at the time of the survey; (2) The number of other businesses owned. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C7: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Income-Generating Activities

VARIABLES \ STATISTICS	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
IGA Count	0.22 (0.00)	0.20 (0.00)	0.04 (0.45)	1.71 (0.95)	(0.72)	(0.00)	(0.00)	2069
Panel B: Components								
Cashcrops and animal husbandry	0.11 (0.01)	0.14 (0.00)	0.04 (0.31)	1.19 (0.80)	(0.59)	(0.10)	(0.03)	2069
Number of other businesses	0.11 (0.00)	0.06 (0.07)	0.00 (0.99)	0.52 (0.57)	(0.22)	(0.00)	(0.06)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are: (1) Whether the participant's household grows crops and/or animals for profits at the time of the survey; (2) The number of other businesses owned. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C8: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Time Use

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Hours worked per week (Wage and self-employed labor)	3.55 (0.00)	6.01 (0.00)	2.91 (0.01)	16.37 (22.95)	(0.06)	(0.60)	(0.02)	2133
Panel B: Components								
Wage labor	-1.65 (0.02)	-0.48 (0.55)	0.62 (0.43)	7.11 (14.63)	(0.14)	(0.00)	(0.19)	2133
Self-employed labor	5.06 (0.00)	6.36 (0.00)	2.27 (0.02)	9.24 (20.37)	(0.28)	(0.01)	(0.00)	2133
Domestic labor	-0.77 (0.28)	-0.70 (0.30)	-0.31 (0.65)	18.07 (14.40)	(0.92)	(0.51)	(0.56)	2133
Unpaid labor	0.03 (0.85)	0.09 (0.65)	-0.18 (0.27)	0.65 (3.12)	(0.78)	(0.21)	(0.13)	2133
Job search	-0.07 (0.00)	-0.04 (0.05)	-0.02 (0.47)	0.08 (0.44)	(0.10)	(0.01)	(0.23)	2133
Attending school	0.03 (0.62)	0.11 (0.12)	-0.00 (0.97)	0.10 (0.89)	(0.27)	(0.61)	(0.12)	2133

Note: Panel A reproduces the results of Table 2. Panel B presents the treatment effect on each component of the index and on other relevant time use variables separately. The dependent variables are the number of hours spent during the 7 days prior to the survey on different activities. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C9: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Time Use

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Hours worked per week (Wage and self-employed labor)	1.43 (0.26)	2.37 (0.06)	2.74 (0.03)	15.58 (24.04)	(0.47)	(0.31)	(0.78)	2069
Panel B: Components								
Wage labor	-0.37 (0.65)	-0.88 (0.26)	0.44 (0.60)	5.16 (14.10)	(0.51)	(0.33)	(0.10)	2069
Self-employed labor	1.72 (0.13)	3.35 (0.00)	2.42 (0.03)	10.36 (20.90)	(0.16)	(0.54)	(0.41)	2069
Domestic labor	-0.31 (0.63)	-0.87 (0.18)	-0.48 (0.47)	16.61 (13.22)	(0.39)	(0.80)	(0.54)	2069
Unpaid labor	0.07 (0.50)	0.11 (0.31)	-0.10 (0.26)	0.29 (1.63)	(0.72)	(0.08)	(0.04)	2069
Job search	0.01 (0.67)	-0.00 (0.87)	-0.00 (0.89)	0.04 (0.33)	(0.55)	(0.56)	(0.98)	2069
Attending school	0.05 (0.22)	0.04 (0.32)	0.00 (0.90)	0.07 (0.60)	(0.84)	(0.28)	(0.39)	2069

Note: Panel A reproduces the results of Table 3. Panel B presents the treatment effect on each component of the index and on other relevant time use variables separately. The dependent variables are the number of hours spent during the 7 days prior to the survey on different activities. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C10: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Mental Health

VARIABLES \ STATISTICS	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Mental Health	0.17 (0.00)	0.17 (0.00)	0.09 (0.12)	0.00 (1.00)	(1.00)	(0.19)	(0.20)	2069
Panel B: Components								
Self-esteem	-0.02 (0.77)	0.08 (0.24)	0.08 (0.16)	0.00 (1.00)	(0.15)	(0.09)	(0.99)	2069
External locus of control	-0.00 (0.96)	0.11 (0.08)	-0.01 (0.82)	0.00 (1.00)	(0.07)	(0.85)	(0.05)	2069
Ladder of life (today)	0.18 (0.00)	0.09 (0.14)	0.04 (0.52)	0.00 (1.00)	(0.12)	(0.02)	(0.41)	2069
Ladder of life (in 5 years)	0.19 (0.00)	0.20 (0.00)	0.12 (0.05)	0.00 (1.00)	(0.96)	(0.18)	(0.18)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are: (1) Self-esteem and (2) locus of control of the participants, and (3) their optimism with regards to their life at the time of the survey; (4) Optimism about their life 5 years ahead. All dependent variables are normalized to be mean 0 and standard deviation 1 in the control group at baseline since mental health related questions were only asked at baseline. In column (T0), we present the normalized average and standard deviation (in parentheses) of the control group at baseline. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C11: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Household Expenditures in 1000 UGX

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Household Expenditure	99.56 (0.00)	70.99 (0.00)	13.05 (0.32)	210.57 (262.23)	(0.06)	(0.00)	(0.00)	2133
Panel B: Components								
Food expenditures	3.43 (0.05)	3.50 (0.04)	1.95 (0.26)	12.97 (35.85)	(0.97)	(0.41)	(0.36)	2133
Rent expenditures	7.03 (0.00)	5.39 (0.02)	0.97 (0.63)	26.35 (42.39)	(0.51)	(0.01)	(0.05)	2133
Non-durables expenditures	6.34 (0.01)	6.26 (0.01)	2.23 (0.37)	40.72 (50.22)	(0.98)	(0.11)	(0.10)	2133
Durables expenditures	39.11 (0.00)	38.73 (0.00)	5.46 (0.22)	26.90 (74.76)	(0.96)	(0.00)	(0.00)	2133
Repairs expenditures	28.36 (0.00)	31.42 (0.00)	4.15 (0.34)	24.18 (72.21)	(0.66)	(0.00)	(0.00)	2133
Clothes expenditures	1.51 (0.01)	1.31 (0.02)	0.43 (0.35)	1.10 (7.19)	(0.77)	(0.07)	(0.12)	2133
Farming for subsistence exp.	10.09 (0.00)	12.93 (0.00)	0.34 (0.86)	11.41 (33.23)	(0.39)	(0.00)	(0.00)	2133
School expenditures	21.64 (0.00)	7.30 (0.12)	2.51 (0.61)	39.36 (86.81)	(0.01)	(0.00)	(0.30)	2133
Gift expenditures	3.15 (0.27)	-0.75 (0.79)	-0.17 (0.95)	28.58 (50.41)	(0.15)	(0.22)	(0.83)	2133
Other expenditures	25.90 (0.00)	7.32 (0.14)	3.57 (0.51)	41.70 (94.15)	(0.00)	(0.00)	(0.46)	2133

Note: Panel A reproduces the results of Table 2. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the amount spent over the 3 weeks prior to the survey on different goods and services in 1000 UGX (the average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP.) In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C12: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Household Expenditures in UGX

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Household Expenditure	48.37 (0.00)	35.53 (0.00)	22.80 (0.02)	130.18 (184.45)	(0.28)	(0.02)	(0.23)	2069
Panel B: Components								
Food expenditures	3.81 (0.00)	4.26 (0.00)	2.03 (0.02)	4.58 (12.54)	(0.66)	(0.06)	(0.03)	2069
Rent expenditures	4.67 (0.00)	3.25 (0.05)	4.45 (0.01)	13.33 (27.89)	(0.43)	(0.90)	(0.49)	2069
Non-durables expenditures	4.04 (0.02)	4.62 (0.01)	2.64 (0.08)	21.10 (28.63)	(0.74)	(0.38)	(0.21)	2069
Durables expenditures	8.71 (0.01)	3.91 (0.15)	3.94 (0.14)	15.04 (42.97)	(0.15)	(0.15)	(0.99)	2069
Repairs expenditures	8.24 (0.01)	4.34 (0.10)	4.54 (0.08)	13.72 (39.67)	(0.24)	(0.26)	(0.94)	2069
Clothes expenditures	-0.01 (0.66)	-0.02 (0.54)	-0.03 (0.35)	0.07 (0.55)	(0.87)	(0.62)	(0.73)	2069
Farming for subsistence exp.	2.78 (0.07)	1.95 (0.14)	0.86 (0.51)	6.67 (22.57)	(0.58)	(0.20)	(0.40)	2069
School expenditures	7.86 (0.18)	4.48 (0.43)	6.04 (0.29)	42.75 (98.12)	(0.56)	(0.76)	(0.78)	2069
Gift expenditures	4.74 (0.01)	4.44 (0.01)	2.32 (0.14)	12.87 (25.62)	(0.87)	(0.16)	(0.21)	2069
Other expenditures	8.40 (0.15)	8.50 (0.15)	6.54 (0.25)	43.18 (98.15)	(0.99)	(0.75)	(0.74)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the amount spent over the 3 weeks prior to the survey on different goods and services in 1000 UGX (the average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP). In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C13: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Housing Conditions

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Housing Conditions Index	0.02 (0.64)	-0.01 (0.87)	-0.03 (0.32)	0.09 (1.05)	(0.55)	(0.16)	(0.46)	2069
Panel B: Components								
Rooms per HH member	0.05 (0.10)	-0.01 (0.82)	0.00 (0.90)	0.53 (0.58)	(0.02)	(0.07)	(0.64)	2069
Household is the owner	0.03 (0.17)	0.01 (0.62)	0.01 (0.55)	0.73 (0.44)	(0.39)	(0.41)	(0.94)	2069
Iron roof	0.04 (0.03)	0.01 (0.39)	0.02 (0.32)	0.51 (0.50)	(0.22)	(0.20)	(0.97)	2069
Concrete walls	0.01 (0.70)	-0.01 (0.79)	-0.01 (0.51)	0.40 (0.49)	(0.53)	(0.31)	(0.72)	2069
Cement floor	0.00 (0.87)	-0.02 (0.34)	-0.02 (0.27)	0.38 (0.49)	(0.26)	(0.20)	(0.90)	2069
Household has electricity	0.01 (0.68)	0.01 (0.48)	0.00 (0.85)	0.14 (0.35)	(0.77)	(0.81)	(0.59)	2069
Household has its own latrine	0.02 (0.55)	0.01 (0.77)	0.04 (0.18)	0.52 (0.50)	(0.75)	(0.48)	(0.29)	2069
Water from protected source	0.02 (0.26)	0.03 (0.11)	0.03 (0.05)	0.90 (0.30)	(0.62)	(0.40)	(0.72)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables consist of: (1) The number of rooms in the participant's dwelling at the time of the survey; (2) Whether they own the dwelling. Whether the dwelling has: (3) A roof made of iron or similar material; (4) Concrete walls; (5) Cement floors; (6) Electricity. Whether household members have (7) access to their own latrine, and (8) to a protected source of water. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C14: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Durable Assets

		GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			
		(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	N
STATISTICS	VARIABLES	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results									
Durable Assets Index		0.17 (0.00)	0.10 (0.01)	0.11 (0.01)	-0.12 (0.81)	(0.11)	(0.23)	(0.75)	2069
Panel B: Components									
Number of phones owned		0.12 (0.02)	0.12 (0.03)	0.12 (0.02)	1.23 (0.95)	(0.97)	(0.99)	(0.98)	2069
Number of radios owned		0.10 (0.01)	0.07 (0.05)	0.09 (0.02)	0.67 (0.68)	(0.48)	(0.83)	(0.64)	2069
Number of televisions owned		0.05 (0.01)	0.01 (0.34)	0.01 (0.44)	0.08 (0.29)	(0.04)	(0.04)	(0.91)	2069
Number of bicycles owned		0.06 (0.10)	0.05 (0.13)	0.03 (0.38)	0.64 (0.69)	(0.87)	(0.46)	(0.55)	2069
Number of motorcycles owned		0.01 (0.58)	-0.00 (0.91)	0.01 (0.56)	0.09 (0.29)	(0.50)	(0.98)	(0.49)	2069
Number of cars owned		0.02 (0.01)	0.00 (0.46)	0.01 (0.29)	0.01 (0.08)	(0.03)	(0.16)	(0.55)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables consist of the number of durable goods owned by the participant and other household members at the time of the survey. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C15: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Savings in 1000 UGX

VARIABLES	STATISTICS	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
		(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
		BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results									
Total Saved		73.05 (0.00)	79.73 (0.00)	-0.54 (0.97)	131.49 (339.32)	(0.64)	(0.00)	(0.00)	2133
Panel B: Components									
Bank savings		14.84 (0.02)	4.94 (0.36)	1.12 (0.80)	30.63 (136.45)	(0.16)	(0.03)	(0.45)	2133
SACCO savings		0.10 (0.97)	1.42 (0.57)	3.00 (0.27)	10.06 (53.35)	(0.60)	(0.29)	(0.55)	2133
Village Savings Account		10.12 (0.02)	15.18 (0.00)	-1.84 (0.66)	33.05 (77.81)	(0.29)	(0.01)	(0.00)	2133
Other savings		41.32 (0.00)	55.46 (0.00)	9.50 (0.08)	35.01 (99.65)	(0.06)	(0.00)	(0.00)	2133

Note: Panel A reproduces the results of Table 2. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the total amount saved in 1000 UGX at the time of the survey in: (1) Banks, (2) Savings and Credit Cooperative Organizations (SACCO); (3) Village savings accounts; (4) Informal savings. The average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C16: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Savings

VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	N
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Total Saved	75.78 (0.00)	77.48 (0.00)	24.07 (0.08)	96.33 (231.17)	(0.93)	(0.00)	(0.00)	2069
Panel B: Components								
Bank savings	12.00 (0.04)	11.53 (0.04)	1.34 (0.79)	19.53 (88.83)	(0.94)	(0.07)	(0.06)	2069
SACCO savings	1.76 (0.65)	1.78 (0.64)	3.42 (0.36)	12.68 (65.28)	(1.00)	(0.68)	(0.67)	2069
Village Savings Account	15.67 (0.00)	24.75 (0.00)	6.60 (0.19)	31.87 (83.74)	(0.12)	(0.10)	(0.00)	2069
Other savings	24.15 (0.00)	31.26 (0.00)	10.17 (0.02)	24.50 (64.06)	(0.30)	(0.01)	(0.00)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the total amount saved in 1000 UGX at the time of the survey in: (1) Banks, (2) Savings and Credit Cooperative Organizations (SACCO); (3) Village savings accounts; (4) Informal savings. The average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP. In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C17: OLS Intent to Treat Estimates One Month Post T1/T2 Grants for Borrowing in 1000 UGX

VARIABLES	STATISTICS	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			
		(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	N
		BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results									
Total Owed		-26.77 (0.25)	-40.53 (0.06)	-37.55 (0.05)	221.92 (489.09)	(0.52)	(0.57)	(0.86)	2133
Panel B: Components									
Family		-3.14 (0.69)	-5.75 (0.54)	-1.29 (0.89)	49.00 (158.70)	(0.78)	(0.84)	(0.69)	2133
Banks		0.64 (0.97)	3.80 (0.84)	-7.05 (0.59)	48.93 (308.84)	(0.86)	(0.53)	(0.47)	2133
Savings group		-8.04 (0.28)	-8.62 (0.23)	-3.66 (0.60)	38.09 (158.28)	(0.91)	(0.39)	(0.29)	2133
MFIs		8.77 (0.37)	-1.65 (0.84)	-2.88 (0.61)	20.42 (112.17)	(0.33)	(0.20)	(0.86)	2133
Moneylenders		1.56 (0.43)	-0.61 (0.14)	0.59 (0.51)	0.64 (9.52)	(0.26)	(0.64)	(0.13)	2133
NGOs		-0.70 (0.84)	-3.25 (0.32)	-1.06 (0.72)	6.35 (60.58)	(0.38)	(0.90)	(0.32)	2133
Age advances		4.81 (0.54)	0.74 (0.74)	-2.13 (0.29)	2.33 (43.40)	(0.60)	(0.36)	(0.08)	2133
School fees		-13.43 (0.05)	-9.35 (0.19)	-3.72 (0.63)	29.25 (206.87)	(0.44)	(0.07)	(0.34)	2133
Landlord		-6.24 (0.50)	-5.10 (0.49)	-7.88 (0.27)	15.71 (151.11)	(0.89)	(0.82)	(0.56)	2133
Airtime		-0.57 (0.31)	-0.42 (0.46)	-0.56 (0.32)	0.58 (12.93)	(0.18)	(0.83)	(0.21)	2133
Shopkeepers		-0.48 (0.33)	-0.52 (0.29)	0.05 (0.93)	1.45 (10.20)	(0.87)	(0.16)	(0.10)	2133
In-kind		-1.53 (0.58)	-1.80 (0.53)	-1.99 (0.42)	6.84 (52.17)	(0.91)	(0.80)	(0.93)	2133
Other		-0.53 (0.84)	5.85 (0.49)	-1.19 (0.64)	6.26 (48.11)	(0.45)	(0.77)	(0.40)	2133

Note: Panel A reproduces the results of Table 2. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the total amount borrowed in 1000 UGX at the time of the survey from different sources (the average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP). In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

Table C18: OLS Intent to Treat Estimates One Year Post T1/T2 Grants for Borrowing

STATISTICS VARIABLES	GRANT	GRANT+ PLANNING	GRANT DELAYED 1-YEAR	CONTROL	TESTS OF EQUALITY			N
	(T1)	(T2)	(T3)	(T0)	T1=T2	T1=T3	T2=T3	
	BETA (P-VALUE)	BETA (P-VALUE)	BETA (P-VALUE)	MEAN (SD)	(P-VALUE)	(P-VALUE)	(P-VALUE)	
Panel A: Primary Results								
Total Owed	27.21 (0.35)	9.73 (0.72)	-20.78 (0.40)	249.38 (513.38)	(0.54)	(0.07)	(0.21)	2069
Panel B: Components								
Family	20.54 (0.03)	20.66 (0.11)	18.35 (0.07)	37.51 (120.66)	(0.99)	(0.86)	(0.88)	2069
Banks	-24.26 (0.34)	-15.26 (0.57)	-29.93 (0.22)	76.21 (506.53)	(0.70)	(0.77)	(0.52)	2069
Savings group	-1.52 (0.82)	8.48 (0.17)	1.32 (0.83)	35.61 (105.48)	(0.13)	(0.67)	(0.25)	2069
MFIs	35.31 (0.04)	4.21 (0.74)	18.81 (0.34)	27.14 (199.93)	(0.08)	(0.48)	(0.47)	2069
Moneylenders	1.12 (0.46)	-0.05 (0.97)	-0.44 (0.67)	1.52 (19.12)	(0.47)	(0.26)	(0.75)	2069
NGOs	2.36 (0.06)	0.54 (0.53)	0.37 (0.71)	1.70 (13.25)	(0.15)	(0.14)	(0.86)	2069
Age advances	-0.51 (0.76)	-1.18 (0.43)	-2.44 (0.05)	2.87 (28.07)	(0.65)	(0.14)	(0.15)	2069
School fees	2.33 (0.76)	1.83 (0.81)	-11.51 (0.09)	48.16 (139.05)	(0.95)	(0.03)	(0.06)	2069
Landlord	-5.01 (0.60)	1.23 (0.89)	-7.56 (0.29)	21.76 (137.97)	(0.54)	(0.74)	(0.24)	2069
Airtime	-0.01 (0.77)	0.02 (0.40)	0.02 (0.61)	0.10 (0.42)	(0.22)	(0.43)	(0.85)	2069
Shopkeepers	0.66 (0.63)	-0.76 (0.41)	0.24 (0.84)	3.89 (17.91)	(0.24)	(0.77)	(0.32)	2069
In-kind	8.65 (0.02)	5.93 (0.01)	2.56 (0.08)	1.24 (8.39)	(0.51)	(0.11)	(0.10)	2069
Other	-0.41 (0.80)	-1.10 (0.48)	-1.15 (0.45)	2.76 (28.83)	(0.63)	(0.59)	(0.97)	2069

Note: Panel A reproduces the results of Table 3. Eicker-White p-values are in parentheses. Panel B presents the treatment effect on each component of the index separately (p-values in parentheses). The dependent variables are the total amount borrowed in 1000 UGX at the time of the survey from different sources (the average exchange rate for 2013 was 2584.88 UGX to a USD\$ or 1036.87 UGX to a USD\$ in PPP). In column (T0), we present the average value of the variable in the control group at baseline and its standard deviation in parentheses. We regress the various outcomes of interests on dichotomous variables for whether participants are in the first treatment arm (T1), second treatment arm (T2), or last treatment arm (T3), leaving the control group as the excluded group. We also include strata fixed effects and control for the outcomes at baseline in all regressions. Under the "Test of Equality" columns, we present the heteroskedasticity-robust p-values of t-tests of equality between the coefficients of different treated groups.

D Data Appendix

D.1 Indexes

Food security: This index captures the extent to which household members went to bed hungry, did not eat for complete days, and did not have any food in the house during the 4 weeks prior to the survey. In particular, the following questions were asked:

- In the past 4 weeks, how often was there no food to eat of any kind in your house because of lack of money or resources to get food?
- In the past 4 weeks, how often did you or any household member go to sleep at night hungry because there was not enough food?
- In the past 4 weeks, how often did you or any household member go a whole day and night without eating anything at all because there was not enough food?

For each question, the individuals chose one of the four following answer choices: never, rarely (1-2 times during the last 4 weeks), sometimes (3-10 times), and often (more than 10 times). The score of the choices are 0,-1,-2, and -3, respectively. In the main analysis, we sum the scores across all 3 questions. Then, we normalize the total score to be mean 0 and standard deviation 1 in the control group at baseline.

Dietary diversity: This variable captures whether different foods were consumed in the 7 days prior to the survey. The following food items were included: (1) Cereals; grains and cereal products such as flours; (2) Roots, tubers and matooke bananas; (3) Nuts and pulses/legumes; (4) Vegetables; (5) Meat, fish and animal products such as eggs and dried meat; (6) Fruits; (7) Milk and dairy products; (8) Fats and oil; (9) Sugars, honey, and sugar products such as jams and sweets; (10) Spices including salt, condiments, and beverages. We asked if the food items were consumed (score=1) or not (score=0). In the main analysis, we sum the scores across all 10 food items. Then, we normalize the total score to be mean 0 and standard deviation 1 in the control group at baseline. In the disaggregated results in appendix, we show the number of days food items were consumed rather than the indices indicating whether the items were consumed, as it provides a more interesting nuance.

Income-Generating Activities: This index essentially captures the number of sources of income of the participants' household. We asked whether their household: (1) Grows crops for

profits; (2) Raises animals for profits; (3) Number of other business(es) owned. The IGA count is the sum of these 3 variables.

Housing Conditions: This variable encompasses the dwelling quality of the participants and their household. To capture the quality of the participant's dwelling, we ask (1) The number of rooms there are in the dwelling that we divide by the number of members; (2) Whether the household owns the dwelling; (3) Whether the dwelling has a roof made of iron or similar sturdy material; (4) Whether it has concrete walls, (5) concrete floors, and (6) electricity; (7) Whether the household has its own latrine and (8) has access to water from a protected source such as a protected well or a protected spring. To construct the index, we first do a Principal Component Analysis (PCA) at baseline and define the raw index as the first component (see Filmer and Pritchett (2001)). At endline, we compute the raw index using the loadings of the first component at baseline. Then, we normalize the raw index to be mean 0 and standard deviation 1 in the control group at baseline.

Durable Assets: This index captures the asset ownership of the participant and their household. We ask how many: (1) Phones, (2) radios, and (3) televisions their household owns. We also ask how many: (4) Bicycles, (5) motorcycles, and (6) cars they possess. We perform the same PCA procedure as above to define index.

Mental Health: We measure the participant's: (1) Self-esteem using Rosenberg's 5-point scale; (2) Rotter's locus of control which measures the participant's beliefs in being able to influence the events that happen in their lives; (3) Optimism today, and (4) optimism in 5 years. Optimism is measured using Cantril's ladder where participants are asked where they see their life today and in 5 years on a scale of 0-10 with 10 being the best possible life for them. Questions related to mental health were only asked at endline. The mental health index is obtained by doing a PCA at endline and normalized to be mean 0 and standard deviation 1 in the control group at endline since the mental health questions were only asked then.

D.2 Continuous variables

Work hours time use: We asked the participants how they divided their active hours during the week (7 days) prior to each survey. We asked about hours spent: (1) On domestic tasks; (2) Working in their own business(es) (self-employed work); (3) Working for pay other than self-employed work; (4) Volunteering; (5) Looking for work; (6) Attending school; (7) Doing other activities excluding

leisure times. We define total work hours as the sum of self-employed work hours and hours worked for pay.²³

Household expenditures: In this module, we asked about household expenditures in the three weeks prior to the survey. We asked about expenditure on: (1) Food; (2) Rent; (3) Non-durables such as toilet paper, cleaning products and personal care items; (4) Durables such as bicycles, cars, furniture and appliances; (5) Clothes; (6) School fees and related expenditures; (7) Gifts and donations; (8) Equipment, inputs, and other expenditures related to subsistence farming; (9) Other expenditures. We sum the expenditures across all 9 categories to obtain the aggregate household expenditure.

Business expenditures: The broad category of business expenditures captures the amount spent on the household's business(es) in the three weeks prior to the survey. This excludes expenditures in subsistence farming and other subsistence activities. Participants were asked how much was spent on: (1) Rent for land and buildings; (2) Inputs of production such as fertilizer and other intermediary inputs; (3) New equipment such as tools, machines, and buildings; (4) Maintenance or repair of equipment; (5) Transportation of products, self and employees; (6) Salaries, wages and compensations to employees, excluding own pay; (7) Purchase of inventory for resell; (8) Other expenditures. We sum the expenditures across all 8 categories to obtain aggregate business expenditures. We also asked questions about business revenues and profits in this module.

Saving and Borrowing: We inquired about the participant's savings by asking how much total savings they had at the time of the survey: (1) In banks; (2) In Savings and Credit Cooperative Organizations (SACCO); (3) In village savings account; (4) In informal savings. The sum of all 4 categories, of which the first 3 are formal saving components, represent the overall amount saved.

In terms of borrowing, we ask the total amount owed at the time of the survey to different individuals and institutions. The borrowing sources considered are: (1) Family members; (2) Banks; (3) SACCO and village groups; (4) Microfinance institutions (MFIs); (5) Moneylenders; (6) NGOs; (7) Wage advances; (8) Advances on school fees; (9) Advances from landlords; (10) Phone airtime loans or advances; (11) Advances or credit from shopkeepers; (12) Value of in-kind debt; (13) Other debts. Overall borrowing is the sum of the amount owed across all categories.

²³In the main analysis, we winsorize the top and bottom 1% of values of total work hours, total expenditures, total savings and borrowing to account for outliers. When looking at individual components of these aggregates, we use the winsorized values of these components, also winsorized at the top and bottom 1%.